
Heterogeneous Judge-Aware Ranking with Reliability, Preference, and Confidence

Anonymous Author(s)

Affiliation

Address

email

Abstract

Pairwise ranking from multiple judges is increasingly central to modern large language model evaluation and preference modeling, yet most existing pipelines still aggregate judgments as if all judges were interchangeable. We develop a *Heterogeneous Judge-Aware (HJA) ranking* framework that separates three features that are often conflated in pooled ranking: judge reliability, structured preference heterogeneity, and statistical confidence. The model yields a consensus ranking shared across judges, while also allowing individual judges to differ both in how strongly they track that consensus and in the systematic ways they deviate from it.

We establish interpretable identifiability conditions for this decomposition, propose a constrained maximum-likelihood estimator, and study estimation in an asymptotic regime tailored to repeated comparisons, where the number of judges may remain fixed or small while the number of comparisons per judge-item pair grows. This regime supports uncertainty quantification not only for the consensus ranking, but also for judge-specific rankings, reliability parameters, and latent disagreement structure.

Empirically, we evaluate the method on synthetic experiments designed to test recovery of consensus, judge heterogeneity, and uncertainty under near-tie settings, and we include a real-data analysis on [\[TOADD\]](#) to illustrate how judge-aware modeling can change substantive ranking conclusions relative to naive pooling. Together, the results provide a statistically grounded framework for ranking under heterogeneous comparative judgments.

1 Introduction

Comparative judgments have become a standard interface for both evaluation and alignment in modern language models. Pairwise preferences are now used not only to build leaderboards and benchmark systems, but also to collect supervision for post-training and preference optimization [1–6]. In these settings, however, the relevant data no longer come from a single homogeneous evaluator. They are increasingly produced by multiple annotators, multiple prompting conditions, or multiple large language models (LLMs), often on the same set of candidate outputs [7–10]. Treating those judges as interchangeable can be seriously misleading: recent evidence shows substantial variability across judge models, languages, and evaluation protocols, and such variability is especially consequential on subjective tasks and near-tie comparisons, where small differences in judging can induce large changes in the final ranking [11–14]. As a result, moving from one judge to many does not merely add noise to a classical ranking problem; it changes the object of inference itself.

The statistical tension is straightforward. Leaderboards are unstable near ties, automatic judges exhibit prompt and language sensitivity, and forcing all judges into one pooled model risks making the final ranking biased rather than more reliable [11–15]. The same issue matters for alignment: preference-based training pipelines treat noisy comparative judgments as supervision, so misspecified judge aggregation can contaminate both evaluation and downstream optimization [4–6, 16].

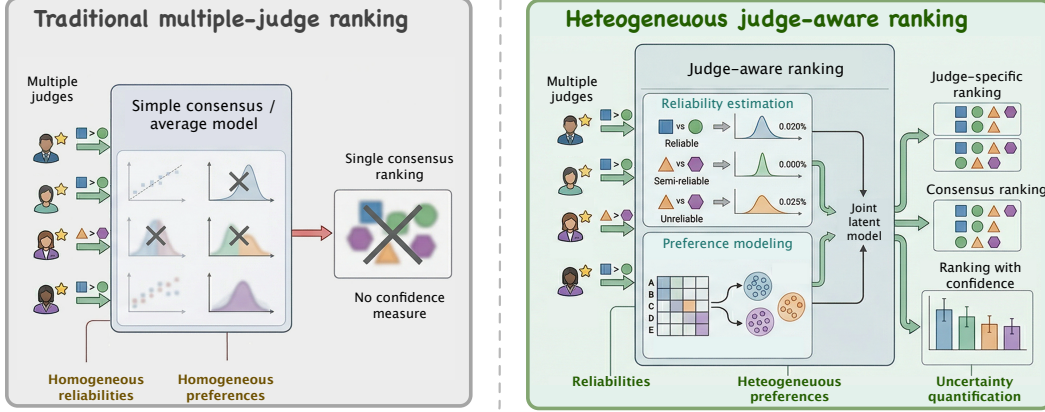


Figure 1: System-level overview of Heterogeneous Judge-Aware (HJA) ranking. Unlike a pooled multi-judge ranking pipeline, which collapses heterogeneous pairwise judgments into a single consensus score, HJA models a judge-specific latent score matrix through the decomposition $S = \gamma\mu^\top + UV^\top$. Here μ represents the consensus ranking direction, γ measures judge-specific sensitivity to that consensus, and $UV^\top$ captures structured preference heterogeneity across judges and items. The resulting procedure outputs consensus and judge-specific rankings together with uncertainty quantification for ranking-relevant contrasts, rather than reporting only point estimates.

To address this gap, we develop a judge-aware ranking framework for repeated pairwise comparisons that separates the main components of comparative evaluation, as summarized in Figure 1. The first is a consensus ranking that captures the ordering most strongly supported across judges. The second is judge-specific reliability, which measures how strongly each judge tracks that shared signal. The third is a structured residual component that captures systematic departures from consensus, allowing judges to differ not only in how noisy or sharp they are, but also in the kinds of items they tend to favor. The structural model of the framework yields an interpretable and estimable representation of heterogeneity. This perspective also preserves uncertainty quantification for the inferential targets that matter in practice, including the consensus ranking, judge-specific rankings, reliability parameters, and latent disagreement patterns.

The technical challenge is therefore not only to posit a flexible model of heterogeneous judging, but to make its decomposition statistically well-defined and inferentially usable. In this paper, we show that the proposed separation of consensus, judge reliability, and structured disagreement is identifiable under explicit normalization and concordance conditions (Proposition 1). We then formulate estimation as a constrained maximum-likelihood problem for repeated pairwise comparisons and develop an anchored proximal alternating procedure (Algorithm 1) with linear convergence guarantee (Theorem 2). Theoretically, we study an asymptotic regime tailored to modern evaluation settings, in which the number of judges may remain fixed or modest while information accumulates through repeated comparisons (Theorem 3). This yields uncertainty quantification not only for the consensus ranking, but also for judge-specific rankings, reliability parameters, and the latent disagreement structure. Taken together, these results provide a statistically grounded alternative to naive pooling for LLM evaluation and preference modeling.

[TOADD] Empirically, ...

2 Related Work

Pairwise ranking, annotator heterogeneity, and judge-aware aggregation. Our starting point is the Bradley–Terry–Luce (BTL) model for ranking from pairwise comparisons [1, 2]. Beyond the homogeneous BTL setting, prior work in crowdsourcing and machine translation evaluation shows that once comparisons are collected from multiple annotators, aggregation must account for annotator-specific reliability, calibration, or sharpness [7, 8]. Our work builds on this line of research but targets a different source of heterogeneity: in LLM-based evaluation, disagreement across judges is often systematic and structured, not merely incidental noise around a single shared ranking. Accordingly, we study a decomposition that separates a consensus ranking direction from judge-specific sensitivity to that direction and from a low-rank residual component capturing structured disagreement.

LLM-as-a-judge and evaluator disagreement. Pairwise LLM judging became a practical standard through benchmarks and platforms such as MT-Bench and Chatbot Arena [3], and early evidence suggested that strong LLM judges could sometimes serve as useful proxies for human evaluation [17]. More recent work has weakened that optimistic view. Large-scale empirical studies report substantial variation across tasks and benchmarks [11]; multilingual evaluations show poor cross-language stability [12]; and rating-indeterminacy analyses argue that collapsing multiple defensible judgments into a single forced label can itself introduce evaluation bias [13]. Near-tie settings are especially fragile, since small prompt or decoding changes can induce large changes in pairwise outcomes and hence in final leaderboards [14]. These observations suggest that disagreement among judges should be modeled as part of the signal, rather than averaged away as if all judges were interchangeable.

Judge-aware ranking and uncertainty quantification. The judge-aware ranking framework JABTL [15] introduces judge-specific discrimination parameters for ranking LLM outputs without ground truth to quantify reliability. Beyond reliability, a related line of work models heterogeneous preferences through structured score matrices learned from pairwise data, especially in collaborative ranking and preference completion [18, 19]. More broadly, low-rank structure has also been studied at the level of pairwise preference probabilities [20], while alternative heterogeneous-population models include user-specific noise or latent user types [21, 22]. Further, related work on comparative uncertainty models argues that evaluation pipelines should retain confidence information rather than reduce judgments to deterministic verdicts [23, 24]. In contrast, we aim to connect these different perspectives to an identifiable, structured latent model for consensus ranking and judge-specific ranking, with reliability, heritability, and confidence.

Protocol design and alignment from comparative feedback. Our paper is also related to work showing that the evaluation protocol itself shapes the inferential target. Ranking without ground truth via triplet comparisons [10] and panel-based judging for subjective tasks [9] both move beyond single-pooled pairwise voting and highlight that multiple defensible judgments may coexist. On the alignment side, comparative judgments are a core supervision source in preference-based learning pipelines [4–6], while recent robustness results emphasize the role of modeling assumptions on both judgments and policies [16].

3 Problem Setup

add some existing references to the following two subsections

3.1 Pairwise judgments from multiple judges

We consider a repeated pairwise comparison setting with multiple judges. Let $[N] = \{1, \dots, N\}$ index the items and $[K] = \{1, \dots, K\}$ index the judges. For each judge $k \in [K]$ and unordered item pair (i, j) with $1 \leq i < j \leq N$, n_{kij} denotes the number of repeated comparisons between items i and j , and $Y_{kij} \in \{0, \dots, n_{kij}\}$ denotes the number of times judge k prefers item i to item j . Write

$$\Omega := \{(k, i, j) : 1 \leq k \leq K, 1 \leq i < j \leq N, n_{kij} > 0\}$$

for the set of observed judge-item-pair triples. This observation scheme is natural in LLM evaluation and preference modeling, where the same comparison may be repeated across prompts, random seeds, or judge models. Throughout the paper, statistical information grows through repeated judgments on observed triples, while the number of judges K may remain fixed or modest.

A pooled ranking model treats all judges as interchangeable and collapses their responses into a single latent score vector. In multi-judge evaluation, however, this is often too restrictive. Judges may differ not only in how strongly they align with an overall ranking signal, but also in systematic ways that cannot be explained by a common score plus independent noise. These considerations suggest that the target of inference should not be a single pooled ranking alone, but a decomposition that preserves a consensus ranking while allowing structured judge-specific departures from it.

3.2 Consensus ranking with reliability, preference heterogeneity, and confidence

A judge-aware ranking model should do more than output a single pooled ordering of the items. In many evaluation settings, the substantive questions are not only which items rank highest on

average, but also which judges track the shared ranking signal most strongly, where systematic preference disagreement remains, and how confident we should be in the ranking conclusions.

Accordingly, we seek a representation that separates three statistical targets: (i) consensus ranking, which summarizes the ordering most strongly supported across judges; (ii) judge-specific reliability, which measures how strongly each judge responds to the shared consensus direction; and (iii) structured preference heterogeneity, which allows judges to differ not only in sharpness or noise level, but also in systematic item-level preferences that cannot be explained by a scalar rescaling alone.

In addition to recovering these latent components, the model should support confidence statements for the inferential outputs they induce. In particular, we would like to quantify uncertainty in the consensus ranking, in judge-specific comparisons, and in the estimated reliability and disagreement structure. This is especially important in near-tie regimes, where small perturbations in judgments can produce large changes in the final ordering. These considerations motivate a structured latent-score model that separates shared ranking signal from judge-specific alignment and residual disagreement. The next section introduces such a decomposition.

4 Heterogeneous Judge-Aware Ranking Model

4.1 Judge-specific latent scores under pairwise comparisons

We model each judge as having a latent score vector over the N items, with pairwise judgments determined by differences in those scores. Let $S \in \mathbb{R}^{K \times N}$ denote the score matrix where the k th row $S_{k,:}$ represents the score vector for judge k . Conditional on S , the counts $\{Y_{kij} : (k, i, j) \in \Omega\}$ are assumed to be independent and follow Bradley–Terry–Luce (BTL) model [1, 25]:

$$Y_{kij} \sim \text{Binomial}(n_{kij}, p_{kij}), \quad \text{logit}(p_{kij}) = S_{ki} - S_{kj}, \quad (k, i, j) \in \Omega. \quad (1)$$

Equivalently, $p_{kij} = \exp(S_{ki}) / \{\exp(S_{ki}) + \exp(S_{kj})\}$. For each judge k , let $\mathcal{G}_k = ([N], \mathcal{E}_k)$ denote the comparison graph with edge set $\mathcal{E}_k = \{\{i, j\} : (k, i, j) \in \Omega\}$. For the main development, we assume that $\mathcal{G}_k$ is connected for every k , as is standard in BTL estimation [2]. This condition implies that each judge-specific score vector is identifiable up to an additive constant from that judge’s observed comparisons. Together with the row-centering constraint

$$S1_N = 0_K, \quad (2)$$

this removes the usual additive invariance of the BTL model, so the collection $\{p_{kij} : (k, i, j) \in \Omega\}$ uniquely identifies S . We allow for disconnected individual comparison graphs in the partial sampling regime of Section C.5, provided that the pooled comparison structure together with the structural constraints of our model still identifies the latent score matrix.

4.2 Consensus-plus-heterogeneity decomposition

To separate shared ranking information from structured disagreement across judges, we model the latent score matrix through a consensus-plus-heterogeneity decomposition:

$$S = \gamma\mu^\top + UV^\top, \quad (3)$$

where $\mu \in \mathbb{R}^N$ is the consensus item score vector, $\gamma \in \mathbb{R}^K$ collects judge-specific consensus-sensitivity parameters, $U \in \mathbb{R}^{K \times r}$ contains judge loadings on latent disagreement directions, and $V \in \mathbb{R}^{N \times r}$ contains the corresponding item coordinates. Equivalently, one has

$$\text{logit}(p_{kij}) = \gamma_k(\mu_i - \mu_j) + U_{k,:}(V_{i,:} - V_{j,:})^\top, \quad (k, i, j) \in \Omega. \quad (4)$$

This decomposition separates two distinct sources of variation across judges. The term $\gamma_k(\mu_i - \mu_j)$ captures how strongly judge k responds to the common consensus signal, and therefore determines how sharply judge k discriminates along the shared ranking direction μ . In the unlikely case where $\gamma_k < 0$, the judge k is viewed as *malicious* [7, 26]. The residual term $U_{k,:}(V_{i,:} - V_{j,:})^\top$ captures systematic departures from that consensus that cannot be explained by a scalar rescaling alone. As a result, the model distinguishes judges who differ mainly in sharpness from judges who exhibit structured preference heterogeneity across items.

This distinction is particularly useful in LLM-as-a-judge settings. Some judges may be weakly aligned with the common ordering and therefore behave as comparatively noisy evaluators, while

others may display systematic deviations tied to answer style, language, task domain, or other latent attributes. Under (3), the consensus vector μ represents the population-level ranking target, whereas $UV^\top$ encodes interpretable disagreement structure around that target.

Remark 1 (Consensus sensitivity versus global reliability). Under (3), the coefficient γ_k modulates only judge k 's response to the consensus direction μ , rather than scaling the entire latent score vector $S_{k,:}$. A judge may be weakly aligned with the common ranking signal while still exhibiting strong and structured idiosyncratic preferences through the heterogeneous component $U_{k,:}V^\top$. In this sense, γ is interpreted as a *consensus-sensitivity* parameter rather than a global reliability parameter.

If one instead wishes γ to scale the full judge-specific score, one may set $U = \text{diag}(\gamma)\tilde{U}$ and write

$$S = \gamma\mu^\top + \text{diag}(\gamma)\tilde{U}V^\top = \text{diag}(\gamma)(1_K\mu^\top + \tilde{U}V^\top).$$

Under this alternative parameterization, γ plays the role of a global reliability parameter, since it scales both the consensus and heterogeneous components simultaneously.

Our formulation does not impose strict positivity on the coefficients γ_k . This is useful in settings with noisy or partially adversarial judges: $\gamma_k = 0$ means that judge k does not respond to the consensus direction and contributes only through heterogeneous preferences, whereas $\gamma_k < 0$ indicates systematic opposition to the consensus signal. By contrast, JA-BTL [15] imposes a strict positivity restriction on the corresponding discrimination parameters.

4.3 Identification of latent components

The decomposition in (3) introduces interpretable latent components for consensus, judge reliability, and structured heterogeneity. However, this factorization is not unique: different parameter tuples may generate the same score matrix S . We therefore impose normalization conditions that make the decomposition statistically meaningful and prepare the ground for identification.

Condition 1 (Normalization). *The parameters (γ, μ, U, V) satisfy:*

- (i) *Centering:* $1_N^\top \mu = 0$ and $V^\top 1_N = 0_r$.
- (ii) *Scaling:* $1_K^\top U = 0_r^\top$ and $1_K^\top \gamma = K$.
- (iii) *Orthogonality:* $\mu^\top V = 0_r^\top$.
- (iv) *Subspace anchoring:* $N^{-1}V^\top V = I_r$ and $K^{-1}U^\top U = D$ for some diagonal matrix D with strictly decreasing positive diagonal entries.

Condition 1 (i)-(ii) fix a canonical representative of the decomposition in (3). The centering constraints remove the additive indeterminacy inherited from the BTL model and align the decomposition with the row-centering normalization $S1_N = 0_K$. The scaling constraints then calibrate the consensus component so that μ represents the cross-judge consensus direction and γ measures judge-specific alignment with that direction. Indeed, under Condition 1 (ii),

$$\frac{1}{K}1_K^\top S = \frac{1}{K}1_K^\top \gamma\mu^\top + \frac{1}{K}1_K^\top UV^\top = \mu^\top, \quad \gamma = \frac{S\mu}{\|\mu\|_2^2} = \frac{K}{\|1_K^\top S\|_2^2} S S^\top 1_K.$$

Thus μ is identified as the row-average score direction, while γ records how strongly each judge loads on that consensus component.

Condition 1 (iii)-(iv) separate consensus from heterogeneity and anchor the residual factorization. The constraint $\mu^\top V = 0_r^\top$ requires the heterogeneous item directions to be orthogonal to the consensus direction, while the subspace-anchoring condition removes rotational ambiguity up to column-wise sign changes. Define the heterogeneous component of the latent score matrix:

$$\tilde{S} := S\mathcal{P}_{S^\top 1_K}^\perp, \tag{5}$$

where $\mathcal{P}_A^\perp$ denotes the projection matrix onto the orthogonal complement of A 's column space. In the degenerate case $S^\top 1_K = 0_N$, the induced model reduces to pure heterogeneity.

Accordingly, for a fixed rank r , we define

$$\Theta_r = \{(\gamma, \mu, U, V) : \mu \in \mathbb{R}^N, \gamma \in \mathbb{R}^K, U \in \mathbb{R}^{K \times r}, V \in \mathbb{R}^{N \times r}, \text{Condition 1 holds}\}.$$

Necessarily $r \leq R_{\max} := \min\{K - 1, N - 2\}$, so that the centered and anchored factors can have full column rank. While Condition 1 specifies a normalized parameterization, it does not by

itself guarantee that an arbitrary centered score matrix S admits such a decomposition. To recover (γ, μ, U, V) uniquely from S , we therefore impose one additional regularity condition on the heterogeneous component $\tilde{S}$, stated in Condition 2.

Condition 2 (Rank). *The heterogeneous score matrix $\tilde{S}$ has r distinct positive singular values.*

The next proposition formalizes the identifiability of the model. First, the observable comparison law identifies the latent score matrix S , and then the normalized consensus-plus-heterogeneity decomposition can be recovered from S uniquely up to column-wise sign changes in (U, V) .

Proposition 1 (Identification). *Consider the probabilistic preference model (1) subject to (2). Then,*

- (i) *(BTL identification) The latent score matrix S can be identified from the generative probability distribution of pairwise comparisons in (1).*
- (ii) *(Parameter identification and recovery) If Condition 2 holds for S , then there exists $(\gamma, \mu, U, V) \in \Theta_r$ such that (3) holds. Such existence is unique up to some sign changes in the columns of U and V .*

Proposition 1 shows that the model parameters correspond to uniquely defined features of the data-generating mechanism, rather than to an arbitrary factorization of S . The only residual ambiguity is the usual column-wise sign indeterminacy in (U, V) , which leaves both $UV^\top$ and the induced pairwise probabilities unchanged. This identification result provides the foundation for the constrained estimation and uncertainty quantification procedures developed in the next section.

5 Estimation and Uncertainty Quantification

5.1 Constrained maximum likelihood estimation

We first consider estimation for a fixed heterogeneity rank $r \leq R_{\max} := \min\{K - 1, N - 2\}$. Let $\theta = (\gamma, \mu, U, V) \in \Theta_r$. For each observed triple $(k, i, j) \in \Omega$, define $\eta_{kij}(\theta) = \gamma_k(\mu_i - \mu_j) + U_{k,:}(V_{i,:} - V_{j,:})^\top$. Dropping constants independent of θ , the negative log-likelihood is

$$\mathcal{L}_n(\theta) := \sum_{(k,i,j) \in \Omega} n_{kij} \left[-\bar{Y}_{kij} \eta_{kij}(\theta) + \log \{1 + \exp(\eta_{kij}(\theta))\} \right], \quad (6)$$

where $\bar{Y}_{kij} = Y_{kij}/n_{kij}$. We estimate the identifiable components of the HJA model by the constrained maximum-likelihood estimator (MLE)

$$\hat{\theta} = (\hat{\gamma}, \hat{\mu}, \hat{U}, \hat{V}) \in \arg \min_{\theta \in \Theta_r} \mathcal{L}_n(\theta). \quad (7)$$

The constraint set Θ_r is not a generic regularizer: it encodes the normalizations used in Section 4.3 to make the consensus direction, judge-specific consensus sensitivities, and low-rank disagreement component statistically identifiable. Thus $\hat{\theta}$ estimates the canonical representative of the decomposition $S = \gamma\mu^\top + UV^\top$, rather than an arbitrary factorization of the same score matrix. The resulting optimization problem is nonconvex because of the bilinear structure in (U, V) and the nonlinear anchoring constraints. The next subsection describes a proximal alternating algorithm that exploits the blockwise smoothness of $\mathcal{L}_n$ while preserving the identifying geometry of Θ_r .

5.2 Anchored alternating optimization

The constrained objective in (7) is nonconvex because of the bilinear factorization $S = \gamma\mu^\top + UV^\top$ and the nonlinear anchoring constraints in Θ_r . Nevertheless, when either the item block (μ, V) or the judge block (γ, U) is fixed, the linear predictor $\eta_{kij}(\theta)$ is affine in the remaining block. Hence, the logistic loss is blockwise convex, and adding proximal terms makes the two subproblems strongly convex. This motivates the alternating updates in Algorithm 1 over the judge and item blocks.

A direct alternating scheme, however, does not preserve the nonlinear identification constraints, including $\mu^\top V = 0$, $N^{-1}V^\top V = I_r$, and the diagonal normalization of $K^{-1}U^\top U$. Algorithm 1 therefore combines proximal block updates with an S -preserving re-anchoring step. This step leaves the induced score matrix unchanged while replacing the current factorization by its canonical representative in Θ_r , thereby maintaining the geometry required by Proposition 1. Further implementation details, including initialization and the re-anchoring routine, are given in Section B.1.

Algorithm 1 Proximal anchored alternating MLE for heterogeneous judge-aware (HJA) ranking

Require: Data $\{(n_{kij}, Y_{kij})\}_{(k,i,j) \in \Omega}$, rank r , tolerance ε , iterations T , and parameters $\{\tau^{(t)}\}_{t \geq 0}$.
Ensure: Estimated parameter $\hat{\theta}$, Fisher information matrix $\mathcal{I}_n(\hat{\theta})$, and a smooth target estimand q .

- 1: $\theta^{(0)} \leftarrow \text{INITIALIZE}(\{(n_{kij}, Y_{kij})\}_{(k,i,j) \in \Omega}, r)$ based on Algorithm B.1.
- 2: **for** $t = 0, 1, \dots, T - 1$ **do**
- 3: Set $\tau \leftarrow \tau^{(t)}$.
- 4: Compute $(\tilde{\gamma}, \tilde{U})$ as the unique minimizer of $\mathcal{L}_n(\gamma, \mu^{(t)}, U, V^{(t)}) + \frac{\tau}{2} \|\gamma - \gamma^{(t)}\|_2^2 + \frac{\tau}{2} \|U - U^{(t)}\|_F^2$ subject to $1_K^\top \gamma = K$ and $1_K^\top U = 0_r^\top$.
- 5: Compute $(\tilde{\mu}, \tilde{V})$ as the unique minimizer of $\mathcal{L}_n(\tilde{\gamma}, \mu, \tilde{U}, V) + \frac{\tau}{2} \|\mu - \mu^{(t)}\|_2^2 + \frac{\tau}{2} \|V - V^{(t)}\|_F^2$ subject to $1_N^\top \mu = 0$ and $1_N^\top V = 0_r^\top$.
- 6: $\theta^{(t+1)} \leftarrow \text{REANCHOR}(\tilde{\gamma}, \tilde{\mu}, \tilde{U}, \tilde{V})$ based on Algorithm B.2.
- 7: **if** $|\mathcal{L}_n(\theta^{(t+1)}) - \mathcal{L}_n(\theta^{(t)})|/[1 + |\mathcal{L}_n(\theta^{(t)})|] < \varepsilon$ **then**
- 8: **break**
- 9: **end if**
- 10: **end for**
- 11: $\hat{\theta} \leftarrow \theta^{(t+1)}$ and $\mathcal{I}_n(\hat{\theta}) \leftarrow \text{ESTIMATEFISHERINFO}(\hat{\theta})$ based on Algorithm B.3.
- 12: Compute $\hat{q} = q(\hat{\theta})$, $\widehat{\text{se}}(\hat{q})$, and the Wald interval $\hat{q} \pm z_{1-\alpha/2} \widehat{\text{se}}(\hat{q})$ using Algorithm B.4.

Unlike JA-BTL [15], we do not log-reparameterize the judge-specific coefficient γ . In our model, γ is an affine-anchored consensus-sensitivity vector recovered from the structured score matrix, not a strictly positive global discrimination parameter. A log transform would impose unnecessary positivity and would not be compatible with the S -preserving re-anchoring map. The proximal terms make the block updates well posed and provide descent, while re-anchoring restores the canonical representative without changing the likelihood.

To analyze Algorithm 1 in the local nonconvex regime, we work in an identified chart around the target constrained MLE. Assumption A1 ensures that the re-anchoring map is locally well defined and nondegenerate; Assumption A2 imposes an isolated constrained local minimizer with positive curvature in this chart; and Assumption A3 places the initialization in its basin of attraction. Let $\hat{\theta}$ be defined by (7), and let $\{\theta^{(t)}\}_{t \geq 0}$ denote the anchored sequence generated by Algorithm 1. Throughout the remainder of Section 5, we fix the local column signs of (U, V) by the anchoring convention and work in the corresponding local identified parameterization.

Theorem 2 (Locally linear convergence of Algorithm 1). *Under Assumptions A1–A3, the iterates remain in the identified local chart, the likelihood values $\{\mathcal{L}_n(\theta^{(t)})\}_{t \geq 0}$ are nonincreasing, and $\theta^{(t)} \rightarrow \hat{\theta}$. Moreover, there exist constants $C < \infty$, $\rho \in (0, 1)$, and $t_0 \geq 0$ such that, for all $t \geq t_0$, $\|\theta^{(t)} - \hat{\theta}\|_2 + \mathcal{L}_n(\theta^{(t)}) - \mathcal{L}_n(\hat{\theta}) \leq C\rho^{t-t_0}$.*

Theorem 2 suggests that the re-anchored alternating updates of Algorithm 1 are stable within the identified local chart: the objective decreases monotonically, the iterates converge to the intended constrained MLE, and the final local convergence is geometric.

5.3 Uncertainty Quantification

Point estimates alone are insufficient for judge-aware ranking: in near-tie regimes, small changes in the observed comparisons can alter the consensus ordering, judge-specific rankings, or estimated disagreement structure. We therefore need uncertainty quantification for the score contrasts that determine ranking conclusions, not only estimates of the latent components themselves. To obtain such uncertainty statements, we study a fixed-rank repeated-comparison asymptotic regime. Unlike classical large-panel factor analysis, the number of judges K and items N may remain fixed, while information accumulates through repeated comparisons on the observed triples. This regime is natural for LLM evaluation, where the same judge panel can be queried repeatedly across prompts, random seeds, or judging instances, even when adding new trusted judges is costly.

We impose the following regularity conditions for statistical inference in this setting. Assumption A4 assumes stable asymptotic sampling fractions $n_{kij}/n \rightarrow w_{kij}$; Assumption A5 requires the active comparison graph for each judge to be connected; Assumption A6 imposes the nondegenerate true structured score model; and Assumption A7 ensures that the structured parameterization is

locally regular after fixing the column signs of (U, V) . Together, graph connectivity identifies the row-centered score matrix from active pairwise differences, while local regularity ensures that this information passes to the consensus, reliability, and heterogeneity parameters.

With the local sign convention fixed as in Section 5.2 and Ω_w defined in (C.1), define the Fisher information for the anchored true parameter $\theta^* \in \Theta_r$ by

$$\mathcal{I}(\theta^*) = \sum_{(k,i,j) \in \Omega_w} w_{kij} p_{kij}^* (1 - p_{kij}^*) \nabla_{\theta} \eta_{kij}(\theta^*) \nabla_{\theta} \eta_{kij}(\theta^*)^{\top}. \quad (8)$$

Under these assumptions, the next theorem establishes consistency and asymptotic normality of the constrained MLE, together with the delta-method limit for smooth target functionals.

Theorem 3 (Fixed-rank asymptotics). *Under Assumptions A4–A7, the constrained MLE satisfies $\hat{\theta} \xrightarrow{P} \theta^*$. Moreover, $\mathcal{I}(\theta^*)$ is positive definite in the identified local parameterization, and*

$$\sqrt{n}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta^*)^{-1}).$$

Consequently, for any smooth scalar target $q : \Theta_r \rightarrow \mathbb{R}$,

$$\sqrt{n}\{q(\hat{\theta}) - q(\theta^*)\} \xrightarrow{d} \mathcal{N}(0, \nabla_{\theta} q(\theta^*)^{\top} \mathcal{I}(\theta^*)^{-1} \nabla_{\theta} q(\theta^*)).$$

Theorem 3 shows that repeated judgments can support inference on the structured components of the HJA model even when the judge panel itself does not grow. The Fisher information in (8) can be consistently estimated by the plug-in observed design estimator

$$\mathcal{I}_n(\hat{\theta}) = \sum_{(k,i,j) \in \Omega} \frac{n_{kij}}{n} \hat{p}_{kij} (1 - \hat{p}_{kij}) \nabla_{\theta} \eta_{kij}(\hat{\theta}) \nabla_{\theta} \eta_{kij}(\hat{\theta})^{\top}, \quad (9)$$

where $\hat{p}_{kij} = \{1 + \exp[-\eta_{kij}(\hat{\theta})]\}^{-1}$. Under the assumptions of Theorem 3, $\mathcal{I}_n(\hat{\theta}) \xrightarrow{P} \mathcal{I}(\theta^*)$. Thus Wald standard errors for smooth targets $q(\theta)$ are given by $\widehat{\text{se}}\{q(\hat{\theta})\} = [\frac{1}{n} \nabla q(\hat{\theta})^{\top} \mathcal{I}_n(\hat{\theta})^{-1} \nabla q(\hat{\theta})]^{1/2}$. Typical choices of q include consensus contrasts $\mu_i - \mu_j$, judge-specific contrasts $S_{ki} - S_{kj}$, consensus-sensitivity parameters γ_k , pairwise probabilities p_{kij} , and smooth summaries of $UV^{\top}$. These are the natural inferential objects for ranking: full rankings are discontinuous at ties, whereas pairwise score differences remain smooth and directly quantify ranking uncertainty.

5.4 Rank selection

The algorithmic and theoretical properties above rely on a fixed model, and we now turn to model selection. The rank r of the factorization $UV^{\top}$ serves as a structural hyperparameter that controls the model’s capacity to capture heterogeneous preferences, thereby affecting model complexity. Section D offers three ways to select r , among which the BIC method has statistical guarantees:

Theorem 4 (Consistency of rank selection). *Under Assumptions A4–A7 with fixed comparison graph, the rank selected by minimizing BIC defined in (D.1) is statistically consistent, i.e.,*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left\{ \arg \min_{r \in \{0, \dots, R_{\max}\}} \text{BIC}(r) = r^* \right\} = 1.$$

6 Experimental Program

The empirical goal is not merely to output another leaderboard. The experiments are designed to reveal when pooled judging fails, when judge-aware modeling helps, and how much ranking conclusions depend on prompt and protocol choices. We organize the evaluation into four blocks, summarized in Table 1.

6.1 Synthetic recovery

The first block tests whether the model can recover the latent consensus ranking and judge-specific reliability under controlled heterogeneity. We simulate μ, γ, U , and V , form $S = \gamma \mu^{\top} + UV^{\top}$, and draw repeated pairwise outcomes from the induced BTL probabilities. This block measures score error, reliability estimation error, top- k coverage, and uncertainty calibration. It is the cleanest setting for separating a genuinely good estimator from one that only looks good when the data are pooled.

Table 1: Summary of extracted experimental table data for Judge-Aware Ranking for LLM Alignment with Reliability and Preference Heterogeneity.

Block	Goal	Data	Judges	Main Outputs
Synthetic recovery	Recover latent consensus and judge reliability under controlled heterogeneity	Simulated pairwise comparisons from the proposed model	Simulated judges with varying γ_k and low-rank disagreement	score error, reliability error, top-k coverage, calibration
Prompt sensitivity	Measure ranking stability under prompt variation	MT-Bench-style question-answer pairs and local pairwise datasets	One judge model under multiple prompts	pairwise flip rate, Kendall correlation, top-k stability
Panel judging	Estimate judge-specific heterogeneity across models/providers	Shared answer pairs on subjective and mixed-difficulty tasks	Multiple LLM judges, including free-tier and user-funded APIs	pooled ranking, judge disagreement, reliability estimates
Near-tie benchmark	Stress-test ranking when candidate answers are almost indistinguishable	Curated near-tie answer pairs	Single and panel judges	tie rate, uncertainty, reordering probability

6.2 Prompt sensitivity

The second block asks how much the final ranking changes when the judging prompt changes but the candidate answers stay fixed. We reuse MT-Bench-style pairwise judging as the baseline protocol [3] and compare it against a tie-aware judge-aware prompt that explicitly discourages verbosity bias and allows a tie verdict when answers are substantively indistinguishable. Every comparison is run in both original and swapped answer order. The resulting flip rate, Kendall rank correlation, and top- k stability quantify how much ranking uncertainty is induced by prompt design alone.

6.3 Panel judging

The third block uses multiple judge models or providers on the same answer pairs. The purpose is to estimate pooled rankings and judge-specific disagreement simultaneously. This protocol is motivated by council-style judging for subjective tasks [9] and by evidence that different LLM judges behave differently across languages and task regimes [11, 12]. The key comparison is between naive pooling and the proposed judge-aware decomposition.

6.4 Near-tie benchmark

The fourth block concentrates on answer pairs that are intentionally hard to separate. These near-tie examples are where automatic rankings become least trustworthy [13, 14]. We construct them by pairing adjacent-quality answers, style-edited variants, and semantically close alternatives with multiple plausible judgments. The primary outputs are tie rate, reordering probability, and the sensitivity of top- k membership to small protocol changes.

6.5 Robustness to self-evaluation bias and its detection

A well-documented phenomenon in automated evaluation is self-evaluation bias (or self-preference), where an LLM judge systematically assigns higher scores to responses generated by itself or models from the same provider family. While this bias poses a significant threat to naive aggregation methods when a judging panel is monopolized by a single model family, its impact is fundamentally mitigated in a diverse, multi-judge setting. Therefore, we select a sufficiently large and diverse panel of judges so that the idiosyncratic self-preference of a few biased judges does not heavily skew the overall consensus ranking. Consequently, the shared consensus μ remains robust, naturally diluting minority biases without external or human-annotated ground-truth rankings.

On the other hand, our formulation can also absorb a judge’s systematic self-preference into the low-rank heterogeneity term $UV^\top$. The heatmap of $UV^\top$ should show bright “hotspots” exactly where a judge prefers the output of its own or its family members. This mechanism ensures that μ retains a high-fidelity representation of consensus quality without the need to completely discard a biased judge.

[TODO] need to demonstrate whether self-preference bias exists or not, and if the bias can be captured and removed through our modeling.

Will include them in the experiment section.

7 Conclusion

We presented a judge-aware ranking framework for LLM alignment evaluation that separates consensus ranking from judge reliability and structured disagreement. The model is motivated by a practical failure mode in current LLM-as-a-judge pipelines: rankings are often reported as if prompts, languages, and judges were interchangeable when they are not. The accompanying experimental program targets exactly those failure modes through synthetic recovery, prompt sensitivity, panel judging, and near-tie stress tests. Combined with a local evaluation runner that makes the protocol reproducible, the resulting paper provides both a statistical model and a concrete path toward more defensible leaderboard claims.

References

- [1] Ralph Allan Bradley and Milton E Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- [2] David R Hunter. Mm algorithms for generalized bradley-terry models. *The annals of statistics*, 32(1):384–406, 2004.
- [3] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.
- [4] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.
- [5] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- [6] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36:53728–53741, 2023.
- [7] Xi Chen, Paul N Bennett, Kevyn Collins-Thompson, and Eric Horvitz. Pairwise ranking aggregation in a crowdsourced setting. In *Proceedings of the sixth ACM international conference on Web search and data mining*, pages 193–202, 2013.
- [8] Naoki Otani, Toshiaki Nakazawa, Daisuke Kawahara, and Sadao Kurohashi. Irt-based aggregation model of crowdsourced pairwise comparison for evaluating machine translations. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 511–520, 2016.
- [9] Justin Zhao, Flor Miriam Plaza-del Arco, and Amanda Cercas Curry. Language model council: Democratically benchmarking foundation models on highly subjective tasks. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 12395–12450, 2025.
- [10] Amit Dhurandhar, Rahul Nair, Moninder Singh, Elizabeth Daly, and Karthikeyan Natesan Ramamurthy. Ranking large language models without ground truth. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2431–2452, 2024.
- [11] Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, et al. Llm instead of human judges? a large scale empirical study across 20 nlp evaluation tasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 238–255, 2025.

- 409 [12] Xiyan Fu and Wei Liu. How reliable is multilingual llm-as-a-judge? In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, 2025.
410
411
- 412 [13] Luke Guerdan, Solon Barocas, Ken Holstein, Hanna Wallach, Steven Wu, and Alexandra Chouldechova. Validating llm-as-a-judge systems under rating indeterminacy. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
413
414
- 415 [14] Mingqi Gao, Yixin Liu, Xinyu Hu, Xiaojun Wan, Jonathan Bragg, and Arman Cohan. Re-evaluating automatic llm system ranking for alignment with human preference. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4605–4629, 2025.
416
417
- 418 [15] Mingyuan Xu, Xinzi Tan, Jiawei Wu, and Doudou Zhou. A judge-aware ranking framework for evaluating large language models without ground truth. *arXiv preprint arXiv:2601.21817*, 2026.
419
420
- 421 [16] Erhan Xu, Kai Ye, Hongyi Zhou, Luhan Zhu, Francesco Quinzan, and Chengchun Shi. Doubly robust alignment for large language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
422
423
- 424 [17] Cheng-Han Chiang and Hung-yi Lee. Can large language models be an alternative to human evaluations? In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15607–15631, 2023.
425
426
- 427 [18] Yu Lu and Sahand N Negahban. Individualized rank aggregation using nuclear norm regularization. In *2015 53rd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 1473–1479. IEEE, 2015.
428
429
- 430 [19] Dohyung Park, Joe Neeman, Jin Zhang, Sujay Sanghavi, and Inderjit Dhillon. Preference completion: Large-scale collaborative ranking from pairwise comparisons. In *International Conference on Machine Learning*, pages 1907–1916. PMLR, 2015.
431
432
- 433 [20] Arun Rajkumar and Shivani Agarwal. When can we rank well from comparisons of $o(n \log(n))$ non-actively chosen pairs? pages 1376–1401, 2016.
434
- 435 [21] Tao Jin, Pan Xu, Quanquan Gu, and Farzad Farnoud. Rank aggregation via heterogeneous thurstone preference models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34 (04), pages 4353–4360, 2020.
436
437
- 438 [22] Rui Wu, Jiaming Xu, Rayadurgam Srikant, Laurent Massoulié, Marc Lelarge, and Bruce Hajek. Clustering and inference from pairwise comparisons. In *Proceedings of the 2015 ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems*, pages 449–450, 2015.
439
440
441
- 442 [23] Yassir Fathullah and Mark Gales. Generalised probabilistic modelling and improved uncertainty estimation in comparative llm-as-a-judge. In *Conference on Uncertainty in Artificial Intelligence*, pages 1266–1288. PMLR, 2025.
443
444
- 445 [24] Victor Wang, Michael JQ Zhang, and Eunsol Choi. Improving LLM-as-a-judge inference with the judgment distribution. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Suzhou, China, November 2025. Association for Computational Linguistics.
446
447
- 448 [25] R Duncan Luce et al. *Individual choice behavior*, volume 4. Wiley New York, 1959.
- 449 [26] Jing Li, Rafal Mantiuk, Junle Wang, Suiyi Ling, and Patrick Le Callet. Hybrid-mst: A hybrid active sampling strategy for pairwise preference aggregation. *Advances in neural information processing systems*, 31, 2018.
450
451

Appendix

This serves as an appendix to the main paper. Below, we provide an outline for the appendix along with a summary of the notation used in the main paper and the appendix.

Organization. The content of the appendix is organized as follows.

Appendix	Content
Section A	A.1 Proof of Proposition 1 on model identification.
Section B	B.1 Algorithms and implementation details. B.2 Proof of Theorem 2 on convergence analysis.
Section C	C.1 Regularity assumptions. C.2 Proof of Theorem 3 (consistency and asymptotic normality). C.3 Auxiliary lemmas used in Section C.2. C.4 Graph-induced nondegeneracy of the Fisher information. C.5 Partial-sampling (pooled-graph treatment) regime.
Section D	D.1 Rank selection criteria. D.2 Proof of Theorem 4 on hyperparameter selection consistency.

Notation. An overview of some general notation used in the main paper and the appendix is as follows.

please add notations used throughout below:

In $\mathbb{R}^d$, the j -th standard basis vector is denoted by e_j and the zero (or all-one) vector is denoted by 0_d (or 1_d) or simply 0 (or 1) if the dimension is clear from the context. The cardinality of a set $\mathcal{S}$ is denoted by $|\mathcal{S}|$. The indicator function is denoted by $\mathbb{1}_{\{\cdot\}}$.

We use “ o ” and “ $\mathcal{O}$ ” to denote the little- o and big- $\mathcal{O}$ notations; “ $o_{\mathbb{P}}$ ” and “ $\mathcal{O}_{\mathbb{P}}$ ” are their probabilistic counterparts. For sequences $\{a_n\}$ and $\{b_n\}$, we write $a_n \lesssim b_n$ if $a_n = \mathcal{O}(b_n)$; and $a_n \asymp b_n$ if $a_n = \mathcal{O}(b_n)$ and $b_n = \mathcal{O}(a_n)$. Convergence almost surely, in probability, and in distribution is denoted by “ $\xrightarrow{\text{a.s.}}$ ”, “ $\xrightarrow{\text{P}}$ ”, and “ $\xrightarrow{\text{d}}$ ”, respectively. When “positivity” property is imposed or stated on a matrix, it means positive definiteness. We use $\text{KL}(p \parallel q)$ to denote the Kullback–Leibler (KL) divergence between two distributions p and q .

A Identification

A.1 Proof of Proposition 1

Proof of Proposition 1. The proof consists of three parts. First for (i), given any set of pairwise probabilities, S is uniquely identified if each of its row is restricted to have zero sum (i.e., Eq. (2)). The second part proves that the mapping induced by (3) from Θ_r to the space of S is injective. Finally, Part (iii) imposes (2) on the space of S , which makes such mapping bijective (one-to-one). Part (iii), combined with Part (ii), establishes the unique existence, which is identification.

Part (i): Identification of BTL score matrix. For each fixed judge k , the pairwise probabilities determine all score differences $S_{ki} - S_{kj}$ through the inverse logit (1). Hence, if $\tilde{S}$ is another matrix generating the same pairwise probabilities, then

$$\tilde{S}_{ki} - \tilde{S}_{kj} = S_{ki} - S_{kj} \quad \text{for all } i, j.$$

Therefore, for each k , the vector $\tilde{S}_{k,:} - S_{k,:}$ is constant across items, i.e., there exists $c_k \in \mathbb{R}$ such that

$$\tilde{S}_{k,:} = S_{k,:} + c_k \mathbf{1}_N^\top.$$

Right-multiplying both sides by $\mathbf{1}_N$ yields

$$\tilde{S}_{k,:} \mathbf{1}_N = S_{k,:} \mathbf{1}_N + c_k \mathbf{1}_N^\top \mathbf{1}_N = S_{k,:} \mathbf{1}_N + N c_k.$$

But it is required that $S \mathbf{1}_N = 0$, with the same applying to $\tilde{S}$, and therefore $c_k = 0$ for all $k \in \{1, \dots, K\}$, i.e., $S = \tilde{S}$.

Part (ii): Identification of latent components. Below, we prove that if $S = \gamma \mu^\top + UV^\top$ with $(\gamma, \mu, U, V) \in \Theta_r$, the parameters are uniquely identified subject to some sign changes to be specified later.

Multiplying both sides of (3) with $\frac{1}{K} \mathbf{1}_K^\top$ yields

$$\frac{1}{K} \mathbf{1}_K^\top S = \frac{1}{K} \mathbf{1}_K^\top \gamma \mu^\top + \frac{1}{K} \mathbf{1}_K^\top UV^\top = \mu^\top, \quad (\text{A.1})$$

so μ is immediately identified by the row average of S .

To further separate the consensus item direction from the heterogeneity item subspace, by Condition 1 (iii),

$$S\mu = \gamma \|\mu\|^2 + U(V^\top \mu) = \gamma \|\mu\|^2 \implies \gamma = \frac{S\mu}{\|\mu\|^2}, \quad (\text{A.2})$$

and thus γ is identified.

We proceed to identify U and V from $\tilde{S} := S - \gamma \mu^\top = UV^\top$. Write $U = [u_1, \dots, u_r]$ and $V = [v_1, \dots, v_r]$. By Condition 1 (iv), the vectors $v_1, \dots, v_r$ are orthonormal after normalization by $\sqrt{N}$, and the columns of U are mutually orthogonal with $\|u_j\|_2^2 = K d_j$. Together with the orthogonality condition (Condition 1 (iii)), this yields the orthogonal decomposition

$$\tilde{S} = \sum_{j=1}^r u_j v_j^\top,$$

where the left vectors $u_1, \dots, u_r$ are mutually orthogonal and the right vectors $v_1, \dots, v_r$ are mutually orthogonal. Therefore, after normalization, this is a singular value decomposition of $\tilde{S}$. Its nonzero singular values are

$$\sigma_j = \|u_j\|_2 \|v_j\|_2 = \sqrt{K N d_j}, \quad j = 1, \dots, r.$$

The strict inequalities $d_1 > \dots > d_r > 0$ imply that $\sigma_1, \dots, \sigma_r$ are pairwise distinct and therefore simple.

Now let $\tilde{S} = \tilde{U} \tilde{V}^\top$ be another representation satisfying the same conditions. The same argument shows that this representation also produces an SVD of $\tilde{S}$ with the same simple nonzero singular

values. By the uniqueness of singular vectors associated with simple singular values, there exist signs $\varepsilon_1, \dots, \varepsilon_r \in \{\pm 1\}$ such that

$$\tilde{u}_j = \varepsilon_j u_j, \quad \tilde{v}_j = \varepsilon_j v_j, \quad j = 1, \dots, r.$$

487 For each j , the pair $(\tilde{u}_j, \tilde{v}_j)$ may be replaced by $(-\tilde{u}_j, -\tilde{v}_j)$ without changing the product $\tilde{u}_j \tilde{v}_j^\top$.
 488 And therefore U and V are identified up to the sign changes in their corresponding columns.

489 **Remark 2** (Sign of factors and loadings). In subsequent analysis where such sign ambiguities also
 490 need ruling out, we may require the first non-zero elements of each column of U to be positive, so
 491 that the signs of U and V are anchored and fixed.

Part (iii): Existence of parameters recovered from S . We are left to show that given any matrix $S \in \mathbb{R}^{K \times N}$ satisfying $S1_N = 0$ and Condition 2, these parameters can indeed be constructed. Let $\mu = \frac{1}{K} S^\top 1_K$, and we have $1_N^\top \mu = 0$. Then,

$$\gamma := \frac{1}{\|\mu\|^2} S \mu = \frac{K}{\|1_K^\top S\|^2} S S^\top 1_K$$

492 satisfies $1_K^\top \gamma = K$, which is the second part of Condition 1 (ii).

493 After recovering the consensus part, the remaining $\tilde{S} = S - \alpha S S^\top 1_K 1_K^\top S$, where $\alpha = 1/\|1_K^\top S\|^2$.
 494 Hence, $\tilde{S} = S \mathcal{P}_{S^\top 1_K}^\perp$ where $\mathcal{P}^\perp$ is the projection matrix onto the orthogonal complement. There-
 495 fore, $1_K^\top \tilde{S} = 1_K^\top S \mathcal{P}_{S^\top 1_K}^\perp = 0^\top$ automatically holds. Take the SVD of $\tilde{S}$ to get $\tilde{S} = P \Sigma R^\top$. Let
 496 $V = \sqrt{N} R$ and $U = \frac{1}{\sqrt{N}} P \Sigma$, and Condition 1 (iv) is satisfied. The rank r is also determined to be
 497 $\text{rank}(\tilde{S})$.

498 To verify the first part of Condition Condition 1 (ii), we left-multiply $1_K^\top$ to the equation $\tilde{S} = UV^\top$
 499 and get $(1_K^\top U) V^\top = 1_K^\top \tilde{S} = 0^\top$, i.e., $V(1_K^\top U)^\top = 0$. But $V \in \mathbb{R}^{N \times r}$ is constructed to have full
 500 column rank, and consequently Condition 1 (ii) must hold.

501 Finally, we study $V^\top \mu$. Note that by (5), $\tilde{S} \mu = \frac{1}{K} S \mathcal{P}_{S^\top 1_K}^\perp 1_K^\top S = 0$, and hence $UV^\top \mu = 0$. But
 502 U is also constructed to have full column rank, and $V^\top \mu$ must be zero. Taking the transpose yields
 503 Condition 1 (iii). A similar argument verifies $V^\top 1_N = 0$.

504 Parts (ii) and (iii) jointly corroborate Proposition 1(ii). □

B Estimation

B.1 Algorithm

This section collects the computational steps for the constrained maximum-likelihood estimator Algorithm 1 introduced in Section 5.1.

B.1.1 Initialization

Algorithm B.1 INITIALIZE: construct $(\gamma^{(0)}, \mu^{(0)}, U^{(0)}, V^{(0)})$

Require: Aggregated counts $\{(n_{kij}, Y_{kij})\}_{(k,i,j) \in \Omega}$, target rank r , feasibility margin $\delta_\gamma > 0$

Ensure: Initial value $\theta^{(0)} = (\gamma^{(0)}, \mu^{(0)}, U^{(0)}, V^{(0)}) \in \Theta_r$

- 1: Fit a pooled centered BTL model to obtain $\mu^{(0)}$.
 - 2: Re-center $\mu^{(0)} \leftarrow \mu^{(0)} - \frac{1}{N}(1_N^\top \mu^{(0)})1_N$.
 - 3: Set $\gamma^{(0)} \leftarrow 1_K$.
 - 4: **for** $k = 1, \dots, K$ **do**
 - 5: Fit a centered judgewise BTL model using judge k 's comparisons to obtain $\tilde{S}_{k,:}^{(0)}$.
 - 6: **end for**
 - 7: Form the initial residual matrix $R^{(0)} \leftarrow \tilde{S}^{(0)} - \gamma^{(0)}(\mu^{(0)})^\top$.
 - 8: Judge-center the residual: $R_c^{(0)} \leftarrow R^{(0)} - \frac{1}{K}1_K1_K^\top R^{(0)}$.
 - 9: Compute the rank- r truncated singular value decomposition $R_c^{(0)} \approx P_r \Sigma_r Q_r^\top$.
 - 10: Set $U^{(0)} \leftarrow P_r \Sigma_r^{1/2}$ and $V^{(0)} \leftarrow Q_r \Sigma_r^{1/2}$.
 - 11: Compute $(\gamma^{(0)}, \mu^{(0)}, U^{(0)}, V^{(0)}) \leftarrow \text{REANCHOR}(\gamma^{(0)}, \mu^{(0)}, U^{(0)}, V^{(0)})$ based on Algorithm B.2.
 - 12: $\gamma_k^{(0)} \leftarrow \max\{\gamma_k^{(0)}, \delta_\gamma\}$.
 - 13: **return** $\theta^{(0)} = (\gamma^{(0)}, \mu^{(0)}, U^{(0)}, V^{(0)})$.
-

B.1.2 Re-anchor and enforce identifiability

Algorithm B.2 REANCHOR: canonical re-anchoring with feasibility check

Require: Tentative $(\tilde{\gamma}, \tilde{\mu}, \tilde{U}, \tilde{V})$ from the two block updates, feasibility margin $\delta_\gamma > 0$, tolerance $\delta_\mu > 0$, spectral tolerance $\delta_\sigma > 0$.

Ensure: Either a feasible anchored point $(\gamma^+, \mu^+, U^+, V^+) \in \Theta_r$ or FAIL.

- 1: Set $\mu^+ \leftarrow \tilde{\mu}$.
 - 2: **if** $\|\mu^+\|_2 < \delta_\mu$ **then**
 - 3: **return** FAIL ▷ Avoid instability incurred on updating V
 - 4: **end if**
 - 5: Compute $a \leftarrow \tilde{V}^\top \mu^+ / \|\mu^+\|_2^2$.
 - 6: Set $\tilde{V} \leftarrow \tilde{V} - \mu^+ a^\top$ and $\gamma^+ \leftarrow \tilde{\gamma} + \tilde{U}a$.
 - 7: Form $\tilde{H} \leftarrow \tilde{U}\tilde{V}^\top$.
 - 8: Compute the thin singular value decomposition $\tilde{H} = P\Sigma Q^\top$, with diagonal entries of Σ arranged in decreasing order.
 - 9: **if** $\sigma_r(\Sigma) < \delta_\sigma$ **then**
 - 10: **return** FAIL ▷ Avoid ill-conditioned matrices caused by a large leap
 - 11: **end if**
 - 12: Set $U^+ \leftarrow P\Sigma/\sqrt{N}$, and $V^+ \leftarrow \sqrt{N}Q$. Flip the corresponding columns of (U^+, V^+) so that the first nonzero entry of each column of U^+ is positive.
 - 13: **return** $(\gamma^+, \mu^+, U^+, V^+)$
-

The re-anchoring step restores the nonlinear identification constraints without changing the implied score matrix $\tilde{S} = \tilde{\gamma}\tilde{\mu}^\top + \tilde{U}\tilde{V}^\top$. Concretely, REANCHOR removes the component of $\tilde{V}$ along $\tilde{\mu}$, compensates for this change through the update $\gamma^+ = \tilde{\gamma} + \tilde{U}a$, and then refactorizes the heterogeneity term by a normalized singular value decomposition. If the candidate loading violates the local

515 nondegeneracy conditions, the routine returns FAIL, and the main algorithm increases the proximal
 516 parameter before recomputing the block updates.

517 In exact arithmetic, the post-anchor candidate automatically satisfies $1_K^\top \gamma^+ = K$ because the judge
 518 block enforces $1_K^\top \tilde{\gamma} = K$ and $1_K^\top \tilde{U} = 0_r^\top$.

519 B.1.3 Confidence intervals for target functionals

Algorithm B.3 ESTIMATEFISHERINFO: plug-in Fisher information

Require: Final estimator $\hat{\theta} = (\hat{\gamma}, \hat{\mu}, \hat{U}, \hat{V})$, data $\{(n_{kij}, Y_{kij})\}_{(k,i,j) \in \Omega}$.

Ensure: Plug-in Fisher information matrix $\mathcal{I}_n(\hat{\theta})$.

- 1: Use the local identified parameterization fixed in Section 5.2; all gradients below are taken in this parameterization.
 - 2: **for** each $(k, i, j) \in \Omega$ **do**
 - 3: Compute $\hat{\eta}_{kij} = \hat{\gamma}_k(\hat{\mu}_i - \hat{\mu}_j) + \hat{U}_{k,:}(\hat{V}_{i,:} - \hat{V}_{j,:})^\top$.
 - 4: Set $\hat{p}_{kij} = \{1 + \exp(-\hat{\eta}_{kij})\}^{-1}$.
 - 5: Compute the local gradient $\hat{g}_{kij} = \nabla_{\theta} \eta_{kij}(\hat{\theta})$.
 - 6: **end for**
 - 7: Set $\mathcal{I}_n(\hat{\theta}) \leftarrow \sum_{(k,i,j) \in \Omega} \frac{n_{kij}}{n} \hat{p}_{kij}(1 - \hat{p}_{kij}) \hat{g}_{kij} \hat{g}_{kij}^\top$.
 - 8: **return** $\mathcal{I}_n(\hat{\theta})$.
-

Algorithm B.4 UNCERTAINTYQUANTIFICATION: delta-method confidence intervals

Require: Final estimator $\hat{\theta}$, plug-in information matrix $\mathcal{I}_n(\hat{\theta})$, target functionals $\mathcal{Q}$, confidence level $1 - \alpha$.

Ensure: Wald confidence intervals for $\{q(\theta) : q \in \mathcal{Q}\}$.

- 1: Use the local identified parameterization fixed in Section 5.2.
 - 2: Set $\hat{\Sigma}_\theta \leftarrow \frac{1}{n} \mathcal{I}_n(\hat{\theta})^{-1}$.
 - 3: **for** each smooth scalar target $q \in \mathcal{Q}$ **do**
 - 4: Compute $\hat{q} = q(\hat{\theta})$.
 - 5: Compute the local gradient $\hat{a}_q = \nabla_{\theta} q(\hat{\theta})$.
 - 6: Compute $\widehat{\text{se}}(\hat{q}) = \left(\hat{a}_q^\top \hat{\Sigma}_\theta \hat{a}_q \right)^{1/2}$.
 - 7: Report $\hat{q} \pm z_{1-\alpha/2} \widehat{\text{se}}(\hat{q})$.
 - 8: **end for**
-

520 B.2 Convergence analysis

521 **Assumption A1** (Local algorithmic chart). There exists a compact set $\Theta_{r,0} \subset \Theta_r$ such that: (i) θ^*
 522 lies in the relative interior of $\Theta_{r,0}$ after fixing the local anchor signs of the columns of U and V ; (ii)
 523 for every $\theta = (\gamma, \mu, U, V) \in \Theta_{r,0}$, one has $\|\mu\|_2 \geq \delta_\mu$, and the nonzero singular values of $UV^\top$ are
 524 separated from each other and from zero by at least δ_σ ; (iii) the re-anchoring map is well-defined
 525 and maps $\Theta_{r,0}$ into itself.

526 **Assumption A2** (Local isolated minimizer). For the empirical criterion $\mathcal{L}_n$, there exists a unique
 527 local minimizer $\hat{\theta}_n \in \Theta_{r,0}$, after fixing the anchor signs, and $\nabla_\xi^2 \mathcal{L}_n(\hat{\theta}_n)$ is positive definite in the
 528 local free-coordinate chart ξ .

529 **Assumption A3** (Initialization in basin). The initialized iterate $\theta^{(0)}$ lies in the basin of attraction of
 530 $\hat{\theta}_n$ within $\Theta_{r,0}$.

531 *Proof of Theorem 2.* The proof of Theorem 2 follows from Proposition B.1. □

532 **Proposition B.1** (Convergence of the proximal anchored alternating algorithm). *Consider the se-*
 533 *quence $\{\theta^{(t)}\}_{t \geq 0}$ generated by Algorithm 1, where each block subproblem includes the proximal*
 534 *terms*

$$\frac{\tau_t}{2} \|\gamma - \gamma^{(t)}\|_2^2 + \frac{\tau_t}{2} \|U - U^{(t)}\|_F^2 \quad \text{and} \quad \frac{\tau_t}{2} \|\mu - \mu^{(t)}\|_2^2 + \frac{\tau_t}{2} \|V - V^{(t)}\|_F^2,$$

with proximal parameters uniformly bounded away from zero ($\tau_t \geq \underline{\tau} > 0$) if τ is allowed to change during different iterations.

Part I: Global descent properties. Suppose the structural assumptions (Assumptions A1 and A6) hold. Then:

1. the sequence $\{\mathcal{L}_n(\theta^{(t)})\}_{t \geq 0}$ is nonincreasing and convergent;
2. the iterates satisfy $\sum_{t=0}^{\infty} \|\theta^{(t+1)} - \theta^{(t)}\|^2 < \infty$.

Part II: Local sequence convergence. Suppose further that Assumptions A2 and A3 concerning the basin hold. Then:

3. the iterates remain in $\Theta_{r,0}$ and the full anchored sequence $\theta^{(t)}$ converges to $\hat{\theta}_n$;
4. equivalently, before anchoring, the factor pair $(U^{(t)}, V^{(t)})$ converges to the equivalence class of $(\hat{U}_n, \hat{V}_n)$ up to the column-sign convention, and after re-anchoring the convergence is ordinary parameter convergence.

Part III: Local linear rate. Following Part II, the convergence is locally linear: there exist constants $C > 0$ and $\rho \in (0, 1)$ such that

$$\|\theta^{(t)} - \hat{\theta}_n\| \leq C\rho^t \quad \text{and} \quad \mathcal{L}_n(\theta^{(t)}) - \mathcal{L}_n(\hat{\theta}_n) \leq C\rho^t$$

for all sufficiently large t .

In particular, whenever the sample criterion admits a unique local minimizer $\hat{\theta}_n$ in the identified chart $\Theta_{r,0}$, the output of Algorithm 1 coincides with that local constrained MLE for all sufficiently accurate runs.

Proof of Proposition B.1. Let the parameter space be partitioned into two blocks: the judge block $\phi_1 = (\gamma, U)$ and the item block $\phi_2 = (\mu, V)$, such that the full parameter vector is $\theta = (\phi_1, \phi_2)$.

Part I: Global Descent Properties. This part relies only on the structural conditions in Assumptions A1 and A6 as the properties of proximal alternating minimization.

Claim 1 (Nonincreasing and convergent objective): For the judge block $\phi_1^{(t+1)}$, the update satisfies:

$$\phi_1^{(t+1)} = \underset{\phi_1}{\operatorname{argmin}} \left\{ \mathcal{L}_n(\phi_1, \phi_2^{(t)}) + \frac{\tau_t}{2} \|\phi_1 - \phi_1^{(t)}\|^2 \right\}.$$

Comparing the value at the minimum $\phi_1^{(t+1)}$ against the previous iterate $\phi_1^{(t)}$ yields the sufficient decrease inequality for the first block:

$$\mathcal{L}_n(\phi_1^{(t+1)}, \phi_2^{(t)}) \leq \mathcal{L}_n(\phi_1^{(t)}, \phi_2^{(t)}) - \frac{\tau_t}{2} \|\phi_1^{(t+1)} - \phi_1^{(t)}\|^2.$$

Applying identical logic to the item block update $\phi_2^{(t+1)}$ yields

$$\mathcal{L}_n(\phi_1^{(t+1)}, \phi_2^{(t+1)}) \leq \mathcal{L}_n(\phi_1^{(t+1)}, \phi_2^{(t)}) - \frac{\tau_t}{2} \|\phi_2^{(t+1)} - \phi_2^{(t)}\|^2.$$

Adding these two inequalities provides the global sufficient decrease condition for a full iteration:

$$\mathcal{L}_n(\theta^{(t+1)}) \leq \mathcal{L}_n(\theta^{(t)}) - \frac{\tau_t}{2} \|\theta^{(t+1)} - \theta^{(t)}\|^2 \leq \mathcal{L}_n(\theta^{(t)}). \quad (\text{B.1})$$

The re-anchoring step (Algorithm B.2 and by Assumption A1) acts purely on the rotational and scaling equivalence classes of the bilinear factorization. Because it is an invariant transformation with respect to S , the objective value $\mathcal{L}_n$ remains unchanged before and after re-anchoring. Therefore, the sequence of objective values $\{\mathcal{L}_n(\theta^{(t)})\}$ is monotonically nonincreasing. Since the negative log-likelihood is bounded from below (by zero), the sequence converges to some limit $L^* \geq 0$.

Claim 2 (Square-summability): Summing the sufficient decrease inequality (B.1) from iteration $t = 0$ to T yields a telescoping sum:

$$\sum_{t=0}^T \frac{\tau_t}{2} \|\theta^{(t+1)} - \theta^{(t)}\|^2 \leq \mathcal{L}_n(\theta^{(0)}) - \mathcal{L}_n(\theta^{(T+1)}) \leq \mathcal{L}_n(\theta^{(0)}).$$

569 Using the strict lower bound $\tau_t \geq \underline{\tau} > 0$ and taking the limit as $T \rightarrow \infty$, we obtain:

$$\sum_{t=0}^{\infty} \|\theta^{(t+1)} - \theta^{(t)}\|^2 \leq \frac{2}{\underline{\tau}} \mathcal{L}_n(\theta^{(0)}) < \infty.$$

570 This finite sum guarantees square-summability of the iterate differences, which also trivially implies
 571 that the step sizes converge to zero: $\|\theta^{(t+1)} - \theta^{(t)}\| \rightarrow 0$ as $t \rightarrow \infty$.

572 **Part II: Local Sequence Convergence.** This part invokes the statistical and initialization condi-
 573 tions in Assumptions A2 and A3.

574 *Claim 3 (Convergence to the local minimizer $\hat{\theta}_n$):* By Assumption A2, $\hat{\theta}_n$ is an isolated local min-
 575 imizer. Therefore, there exists a compact neighborhood $\mathcal{B} \subset \Theta_{r,0}$ around $\hat{\theta}_n$ such that $\hat{\theta}_n$ is the
 576 strictly unique stationary point within $\mathcal{B}$, and $\mathcal{L}_n(\theta) > \mathcal{L}_n(\hat{\theta}_n)$ for all $\theta \in \mathcal{B} \setminus \{\hat{\theta}_n\}$.

577 By Assumption A3, the initialization $\theta^{(0)}$ lies close enough to $\hat{\theta}_n$ such that the connected component
 578 of the sublevel set $\mathcal{S} := \{\theta \in \mathcal{B} \mid \mathcal{L}_n(\theta) \leq \mathcal{L}_n(\theta^{(0)})\}$ is strictly contained in the interior of $\mathcal{B}$, given
 579 the positive definiteness of the Hessian matrix. Because $\mathcal{B}$ is compact and $\mathcal{L}_n$ is continuous, the
 580 sublevel set $\mathcal{S}$ is also compact.

581 By the global descent property established in Part I, $\mathcal{L}_n(\theta^{(t+1)}) \leq \mathcal{L}_n(\theta^{(t)})$ for all t . Thus, the
 582 sequence cannot enter the region where $\mathcal{L}_n(\theta) > \mathcal{L}_n(\theta^{(0)})$, meaning the entire sequence $\{\theta^{(t)}\}_{t=0}^{\infty}$
 583 is strictly confined within the compact set $\mathcal{S}$. The re-anchoring step under Assumption A1 preserves
 584 this containment.

585 Because $\mathcal{S}$ is compact, the sequence $\{\theta^{(t)}\}$ must possess at least one limit point $\theta^* \in \mathcal{S}$ by the
 586 Bolzano-Weierstrass theorem. To characterize any such limit point, we invoke the first-order opti-
 587 mality conditions of the exact proximal block updates. For the judge block,

$$\nabla_{\phi_1} \mathcal{L}_n(\phi_1^{(t+1)}, \phi_2^{(t)}) + \tau_t(\phi_1^{(t+1)} - \phi_1^{(t)}) = 0.$$

588 Taking the norm yields $\|\nabla_{\phi_1} \mathcal{L}_n(\phi_1^{(t+1)}, \phi_2^{(t)})\| = \tau_t \|\phi_1^{(t+1)} - \phi_1^{(t)}\|$. By Part I and the boundedness
 589 of τ_t , the right-hand side converges to zero, and thus the gradient mapping uniformly converges
 590 to zero. Because $\mathcal{L}_n$ is continuously differentiable, any subsequence converging to θ^* satisfies
 591 $\nabla \mathcal{L}_n(\theta^*) = 0$.

592 Therefore, every limit point of the sequence is a stationary point. But $\mathcal{S} \subset \mathcal{B}$, and by construction,
 593 $\hat{\theta}_n$ is the strictly unique stationary point within $\mathcal{B}$. Therefore, $\theta^* = \hat{\theta}_n$ is the unique limit point of
 594 the sequence. Since $\{\theta^{(t)}\}$ is a sequence in a compact set with exactly one limit point, the entire
 595 sequence globally converges to $\hat{\theta}_n$.

596 *Claim 4 (Pre- and post-anchoring equivalence):* Because the joint parameters converge, the se-
 597 quence of latent evaluation scores $S^{(t)} = \gamma^{(t)}(\mu^{(t)})^\top + U^{(t)}(V^{(t)})^\top$ converges to the optimal em-
 598 pirical score matrix $\hat{S}_n$. Before the re-anchoring step is applied, the unanchored factor matrices
 599 $(U^{(t)}, V^{(t)})$ converge to the equivalence class $[\hat{U}_n, \hat{V}_n]$ defined by orthogonal rotations and scale-
 600 sign permutations. The re-anchoring map is a continuous function acting on $\Theta_{r,0}$ that uniquely
 601 identifies a representative from this equivalence class by fixing local anchor signs, and hence guar-
 602 antees the ordinary parameter-wise convergence in the Euclidean space.

603 **Part III: Local linear rate.** This part draws the final conclusion on local convergence rate from
 604 the assumed and established properties of the objective function, the algorithm, and the topological
 605 structures of the global optimum.

606 We restrict the domain of the objective function in $\mathcal{S}$, where $\mathcal{L}$ demonstrates strong convexity. By
 607 Theorem 2 of [1], $\mathcal{L}$ satisfies the Polyak-Łojasiewicz (PL) inequality [2], and by Appendix F of [1],
 608 satisfies the proximal-PL property, and by Appendix G of [1], has a local Kurdyka-Łojasiewicz (KL)
 609 exponent $\theta = 1/2$. The local linear convergence rate hence directly follows from Theorem 3.4 of
 610 [3]. $\square$

C Statistical inference

C.1 Regularity assumptions

We formalize the regularity conditions mentioned in Section 5.3 as the assumptions below.

Assumption A4 (Asymptotic sampling fractions). There exist fixed constants $w_{kij} \geq 0$ such that $n_{kij}/n \xrightarrow{p} w_{kij}$ for all $(k, i, j) \in \Omega$ as $n \rightarrow \infty$.

If n_{kij} is non-random, i.e., the design of the experiment is deterministic, the convergence in probability is reduced to convergence in the ordinary sense. Note that a triple can be observed ($n_{kij} > 0$) but still have asymptotic fraction $w_{kij} = 0$. To account for this in subsequent analysis, we define:

$$\Omega_w := \{(k, i, j) \in \Omega : w_{kij} > 0\} \quad (\text{C.1})$$

as the set of asymptotically active triples.

Assumption A5 (Graph connectivity). For each judge $k \in [K]$, the comparison graph $\mathcal{G}_k^w = ([N], \mathcal{E}_k^w)$ with edge set $\mathcal{E}_k^w = \{\{i, j\} : (k, i, j) \in \Omega_w\}$ is connected, and $0 < p_{kij}^* < 1$ for each active triple $(k, i, j) \in \Omega_w$.

By connectedness, we mean an undirected graph where a path (a sequence of connecting edges) exists between every pair of nodes (items). Therefore, under Assumption A5, all the items assessed by a judge should be linked via one or more comparisons. However, it is not required that all possible pairs are directly compared and this scenario is accounted for via Ω_w .

Assumption A6 (Non-degenerate consensus). The true model (3) holds with the true parameter $\theta^* \in \Theta_r$, and the true consensus direction $\mu^* \neq 0$.

C.2 Proof of Theorem 3

Recall that the structural parameter is estimated by the constrained maximum likelihood estimator

$$\hat{\theta} := (\hat{\gamma}, \hat{\mu}, \hat{U}, \hat{V}) \in \arg \min_{\theta \in \Theta_r} \mathcal{L}(S(\theta)), \quad S(\theta) := \gamma\mu^\top + UV^\top.$$

Since the likelihood depends on θ only through the induced score matrix $S(\theta)$, the proof is most naturally carried out on the centered score space

$$\mathcal{S} := \{S \in \mathbb{R}^{K \times N} : S1_N = 0, \text{ and Condition 2 holds}\}.$$

The signs of the columns of (U, V) can be anchored and fixed via, say, Remark 2, and then Proposition 1 ensures $\mathcal{S} = \{S(\theta) : \theta \in \Theta_r\}$.

Proof of Theorem 3. For $S \in \mathcal{S}$, write

$$\eta_{kij}(S) := S_{ki} - S_{kj}, \quad p_{kij}(S) := \frac{1}{1 + \exp\{-\eta_{kij}(S)\}}.$$

Also write $\bar{Y}_{kij} := Y_{kij}/n_{kij}$. Then the normalized negative log-likelihood, viewed as a function of S , is

$$M_n(S) := \frac{1}{n} \mathcal{L}(S) = \sum_{(k,i,j) \in \Omega} \frac{n_{kij}}{n} \left[-\bar{Y}_{kij} \eta_{kij}(S) + \log(1 + \exp\{\eta_{kij}(S)\}) \right]. \quad (\text{C.2})$$

By the strong law, for each fixed $(k, i, j) \in \Omega$,

$$\bar{Y}_{kij} \xrightarrow{\text{a.s.}} p_{kij}^*,$$

and by Assumption A4,

$$\frac{n_{kij}}{n} \rightarrow w_{kij}.$$

Since Ω is fixed and each summand in (C.2) is continuous in S , it follows that $M_n(S)$ converges uniformly on every compact subset of $\mathcal{S}$ to

$$M(S) := \sum_{(k,i,j) \in \Omega} w_{kij} \left[-p_{kij}^* \eta_{kij}(S) + \log(1 + \exp\{\eta_{kij}(S)\}) \right]. \quad (\text{C.3})$$

642 Let $S^* = S(\theta^*) \in \mathcal{S}$. Then, by the usual Bernoulli Kullback–Leibler identity,

$$M(S) - M(S^*) = \sum_{(k,i,j) \in \Omega} w_{kij} \text{KL}(\text{Bern}(p_{kij}^*) \parallel \text{Bern}(p_{kij}(S))) \geq 0. \quad (\text{C.4})$$

643 Equality holds if and only if $p_{kij}(S) = p_{kij}^*$ for all $(k, i, j) \in \Omega_w$, equivalently,

$$S_{ki} - S_{kj} = S_{ki}^* - S_{kj}^* \quad \text{for all } (k, i, j) \in \Omega_w.$$

644 By Lemma C.1, equality in (C.4) implies $S = S^*$. Hence S^* is the unique minimizer of M on $\mathcal{S}$.

645 We first prove the consistency of the score-space MLE $\hat{S} \in \arg \min_{S \in \mathcal{S}} \mathcal{L}(S)$.

646 Fix any open neighborhood $\mathcal{O}$ of S^* in $\mathcal{S}$. Since M is continuous and uniquely minimized at S^* ,
647 there exists $\delta > 0$ such that

$$\inf_{S \in \mathcal{S} \setminus \mathcal{O}} M(S) \geq M(S^*) + 3\delta. \quad (\text{C.5})$$

648 By Lemma C.2, there exists $R < \infty$ such that

$$\inf_{S \in \mathcal{S}: \|S\|_F > R} M(S) \geq M(S^*) + 4\delta. \quad (\text{C.6})$$

649 Define the truncated score set

$$K_R := \{S \in \mathcal{S} : \|S\|_F \leq R\}.$$

650 Since K_R is closed and bounded in the finite-dimensional Euclidean space $\mathbb{R}^{K \times N}$, it is compact.
651 By uniform convergence on compact subsets,

$$\sup_{S \in K_R} |M_n(S) - M(S)| \xrightarrow{\mathbb{P}} 0.$$

652 Hence, with probability tending to one,

$$\sup_{S \in K_R} |M_n(S) - M(S)| < \delta.$$

653 On this event, for every $S \in K_R$,

$$M_n(S^*) \leq M(S^*) + \delta,$$

654 whereas for every $S \in K_R \setminus \mathcal{O}$, by (C.5),

$$M_n(S) \geq M(S) - \delta \geq M(S^*) + 2\delta.$$

655 Thus no minimizer of M_n over K_R can lie in $K_R \setminus \mathcal{O}$.

656 Further, by the sample coercivity part of Lemma C.2, with probability tending to one,

$$\inf_{S \in \mathcal{S}: \|S\|_F > R} M_n(S) > M(S^*) + 2\delta.$$

657 Hence no global minimizer of M_n over $\mathcal{S}$ can lie outside K_R . Combining the two conclusions, any
658 global minimizer $\hat{S}$ of M_n over $\mathcal{S}$ must belong to $\mathcal{O}$ with probability tending to one, i.e.,

$$\mathbb{P}(\hat{S} \in \mathcal{O}) \rightarrow 1.$$

659 Since $\mathcal{O}$ was arbitrary, we obtain

$$\hat{S} \xrightarrow{\mathbb{P}} S^*.$$

660 We now pass the consistency in the score space into the parameter space after fixing the sign conven-
661 tion of Remark 2. Note that the mapping $S(\theta)$ is continuous. Conversely, the mapping established
662 through (A.1) and (A.2) is continuous and smooth. For the remaining SVD part for $\tilde{S}$, Condition 2
663 guarantees the smoothness of the mapping by SVD perturbation theory [4, 5]. Therefore, the map-
664 ping $S(\cdot)$ is invertible, and both $S(\cdot)$ and $S^{-1}(\cdot)$ is smooth. Continuous Mapping Theorem therefore
665 yields

$$\hat{\theta} \xrightarrow{\mathbb{P}} \theta^*.$$

666 We finally derive asymptotic normality. By the delta method, if $\sqrt{n}(\hat{S} - S^*) \xrightarrow{d} \mathcal{N}(0, \Sigma_S)$ for some
 667 covariance Σ_S , then

$$\sqrt{n}(\hat{\theta} - \theta^*) = \sqrt{n}D\Psi(S^*)(\hat{S} - S^*) + o_p(1) \xrightarrow{d} \mathcal{N}(0, D\Psi(S^*)\Sigma_S D\Psi(S^*)^\top),$$

668 where $\Psi : \mathcal{S} \rightarrow \Theta_r$ is the locally invertible recovery map of Proposition 1, which has been shown
 669 to be smooth, and $D\Psi(S^*)$ is its Jacobian.

670 The limiting covariance in the S -space is $\Sigma_S = \mathcal{I}_S(S^*)^{-1}$, where

$$\mathcal{I}_S(S^*) = \sum_{(k,i,j) \in \Omega_w} w_{kij} p_{kij}^* (1 - p_{kij}^*) \nabla_S \eta_{kij}(S^*) \nabla_S \eta_{kij}(S^*)^\top$$

671 is positive definite by Proposition C.3 (Step 1–2).

672 By the chain rule, $\nabla_{\theta} \eta_{kij}(\theta^*) = D\Psi(S^*)^\top \nabla_S \eta_{kij}(S^*)$, so $D\Psi(S^*)\mathcal{I}_S(S^*)D\Psi(S^*)^\top = \mathcal{I}_{\Theta}(\theta^*)$
 673 given by (8), whose non-degeneracy is analyzed in Section C.4, especially Proposition C.3.

674 Therefore,

$$\sqrt{n}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_{\Theta}(\theta^*)^{-1}).$$

675 This proves the theorem.

676 We now calculate the explicit form of the Fisher Information. For each $(k, i, j) \in \Omega_w$, the score
 677 contribution is

$$\ell_{kij}(\theta) = Y_{kij} \eta_{kij}(\theta) - n_{kij} \log(1 + \exp\{\eta_{kij}(\theta)\}),$$

678 where

$$\eta_{kij}(\theta) = \gamma_k(\mu_i - \mu_j) + U_{k,:}(V_{i,:} - V_{j,:})^\top.$$

679 The per-triple Fisher information is

$$\mathcal{I}_{kij}(\theta^*) = \mathbb{E}[\nabla_{\theta} \log p(Y_{kij} | \theta^*) \nabla_{\theta} \log p(Y_{kij} | \theta^*)^\top] = n_{kij} p_{kij}^* (1 - p_{kij}^*) \nabla_{\theta} \eta_{kij}(\theta^*) \nabla_{\theta} \eta_{kij}(\theta^*)^\top.$$

680 Hence the aggregate normalized Fisher information is

$$\mathcal{I}_n(\theta^*) = \sum_{(k,i,j) \in \Omega_w} \frac{n_{kij}}{n} p_{kij}^* (1 - p_{kij}^*) \nabla_{\theta} \eta_{kij}(\theta^*) \nabla_{\theta} \eta_{kij}(\theta^*)^\top,$$

681 and since $n_{kij}/n \rightarrow w_{kij}$,

$$\mathcal{I}_n(\theta^*) \xrightarrow{p} \mathcal{I}_{\Theta}(\theta^*).$$

682 The gradient blocks are

$$\begin{aligned} \frac{\partial \eta_{kij}}{\partial \gamma_k} &= \mu_i - \mu_j, \\ \frac{\partial \eta_{kij}}{\partial \mu_\ell} &= \gamma_k(\delta_{i\ell} - \delta_{j\ell}), \\ \frac{\partial \eta_{kij}}{\partial U_{k,m}} &= (V_i - V_j)_m, \\ \frac{\partial \eta_{kij}}{\partial V_{\ell,m}} &= U_{k,m}(\delta_{i\ell} - \delta_{j\ell}). \end{aligned}$$

683 □

684 C.3 Auxiliary lemmas

685 Two lemmas used in the proof of Theorem 3 are established below.

686 **Lemma C.1** (Injectivity of active score differences). *Under Assumption A5, if $S, S' \in \mathcal{S}$ satisfy*

$$S_{ki} - S_{kj} = S'_{ki} - S'_{kj} \quad \text{for all } (k, i, j) \in \Omega_w,$$

687 *then $S = S'$.*

688 *Proof of Lemma C.1.* Fix a judge k and let $h_k := S_{k,:} - S'_{k,:} \in \mathbb{R}^N$. The assumption implies

$$h_{ki} - h_{kj} = 0 \quad \text{for every } \{i, j\} \in \mathcal{E}_k^w.$$

689 Because the graph $\mathcal{G}_k^w$ is connected, h_k must be constant across items, say

$$h_k = c_k \mathbf{1}_N.$$

690 Since both S and S' lie in the centered space $\mathcal{S}$, we also have

$$h_k^\top \mathbf{1}_N = 0.$$

691 Therefore $Nc_k = 0$, so $c_k = 0$ and hence $h_k = 0$. This holds for every judge k , and thus $S = S'$. $\square$

693 **Lemma C.2 (Coercivity).** *Under Assumption A5, the population criterion M in (C.3) is coercive on $\mathcal{S}$, i.e.*

$$\|S\|_F \rightarrow \infty, \quad S \in \mathcal{S} \quad \implies \quad M(S) \rightarrow \infty.$$

695 *Moreover, with probability tending to one, the sample criterion M_n in (C.2) is also coercive on $\mathcal{S}$.*

696 *Proof of Lemma C.2.* We first prove the coercivity of the population criterion.

697 For each fixed active triple $(k, i, j) \in \Omega_w$, define

$$\phi_{kij}(x) := -p_{kij}^* x + \log(1 + e^x).$$

698 Because $p_{kij}^* \in (0, 1)$, the function ϕ_{kij} is continuous, nonnegative, and satisfies

$$\phi_{kij}(x) \rightarrow \infty \quad \text{as } |x| \rightarrow \infty.$$

699 Indeed, as $x \rightarrow +\infty$,

$$\phi_{kij}(x) = (1 - p_{kij}^*)x + o(x) \rightarrow \infty,$$

700 while as $x \rightarrow -\infty$,

$$\phi_{kij}(x) = -p_{kij}^* x + o(1) \rightarrow \infty.$$

701 Now fix a judge k and write the k th row of S as

$$s_k := (S_{k1}, \dots, S_{kN})^\top \in \mathbb{R}^N.$$

702 Since $S \in \mathcal{S}$, we have $s_k^\top \mathbf{1}_N = 0$. Because $\mathcal{G}_k^w$ is connected and N is fixed, we have

$$\|s_k\|_2 \leq \sqrt{N} \|s_k\|_1 \leq 2\sqrt{N} \sum_{\{i,j\} \in \mathcal{E}_k^w} |s_{k,i} - s_{k,j}|. \quad (\text{C.7})$$

703 To see this, fix a reference node in $\mathcal{G}_k^w$ and telescope along paths from the reference node to each other node; since the graph is finite and connected, all resulting path lengths are uniformly bounded.

705 Suppose now that $\|S\|_F \rightarrow \infty$ with $S \in \mathcal{S}$. Then for some judge k we must have $\|s_k\|_2 \rightarrow \infty$. By (C.7), this implies that for at least one active edge $\{i, j\} \in \mathcal{E}_k^w$,

$$|S_{ki} - S_{kj}| = |s_{k,i} - s_{k,j}| \rightarrow \infty.$$

707 Since the corresponding summand in $M(S)$ is

$$w_{kij} \phi_{kij}(S_{ki} - S_{kj}),$$

708 with $w_{kij} > 0$, it follows that this term diverges to $+\infty$. All other summands are nonnegative, and therefore

$$M(S) \rightarrow \infty.$$

710 This proves population coercivity.

711 We next prove sample coercivity. On the event

$$\mathcal{A}_n := \{0 < Y_{kij} < n_{kij} \text{ for all } (k, i, j) \in \Omega_w\},$$

712 define

$$\phi_{kij,n}(x) := -\bar{Y}_{kij} x + \log(1 + e^x).$$

713 Whenever $0 < \bar{Y}_{kij} < 1$, the same argument as above shows that

$$\phi_{kij,n}(x) \rightarrow \infty \quad \text{as } |x| \rightarrow \infty.$$

714 Since $p_{kij}^* \in (0, 1)$ and $n_{kij} \rightarrow \infty$ for each active triple, we have

$$\mathbb{P}(Y_{kij} = 0 \text{ or } Y_{kij} = n_{kij}) \rightarrow 0,$$

715 and therefore, by a union bound over the finite set Ω_w ,

$$\mathbb{P}(\mathcal{A}_n) \rightarrow 1.$$

716 On $\mathcal{A}_n$, the same graph argument used for the population criterion shows that

$$\|S\|_F \rightarrow \infty, \quad S \in \mathcal{S} \quad \implies \quad M_n(S) \rightarrow \infty.$$

717 Thus M_n is coercive on $\mathcal{S}$ with probability tending to one. $\square$

718 C.4 Graph-induced nondegeneracy of the Fisher information

719 In Theorem 3, the asymptotic normality of the constrained maximum likelihood estimator was stated
720 under the nonsingularity of the Fisher information matrix

$$I_{\Theta}(\theta^*) = \sum_{(k,i,j) \in \Omega_w} w_{kij} p_{kij}^* (1 - p_{kij}^*) \nabla_{\theta} \eta_{kij}(\theta^*) \nabla_{\theta} \eta_{kij}(\theta^*)^{\top}.$$

721 In this section, we show that such nondegeneracy can be derived from the active comparison graph
722 structure together with a local regularity condition on the structured score map. The argument is
723 inspired by the graph-Laplacian viewpoint for pairwise-comparison Fisher information developed in
724 [6], where positivity of the Fisher information is obtained from graph connectivity (after quotienting
725 out the additive invariance of pairwise-comparison models). In our setting, the proof proceeds in
726 two layers:

- 727 1. establish positive definiteness of the Fisher information in the ambient row-centered score space
728 $\mathcal{S}$ via judge-wise connected active graphs; and
- 729 2. pull this positivity back to the structured parameter space Θ_r through the differential of the
730 map $\Phi(\theta) = S(\theta) = \gamma\mu^{\top} + UV^{\top}$.

731 The second step is needed because our inferential target is not the unrestricted score matrix S , but
732 the constrained latent parameter $\theta = (\gamma, \mu, U, V)$.

733 We first isolate the local regularity condition needed to pass from score-space nondegeneracy to
734 parameter-space nondegeneracy.

735 **Assumption A7** (Local chart regularity of the structured score map). Let $\theta^* = (\gamma^*, \mu^*, U^*, V^*) \in$
736 Θ_r denote the true parameter. After fixing the column signs of (U, V) according to a local sign
737 convention, there exists a neighborhood of θ^* in Θ_r that is parameterized by a smooth local chart on
738 the manifold. Moreover, the differential of the structured score map

$$\Phi : \Theta_r \rightarrow \mathcal{S}, \quad \Phi(\theta) = S(\theta) = \gamma\mu^{\top} + UV^{\top},$$

739 is injective at θ^* on the tangent space $T_{\theta^*}\Theta_r$. That is,

$$D\Phi(\theta^*)[\delta\theta] = 0, \quad \delta\theta \in T_{\theta^*}\Theta_r \quad \implies \quad \delta\theta = 0.$$

740 Assumption A7 is the infinitesimal analog of parameter identifiability at the local chart of θ^* .

741 **Proposition C.3** (Graph-induced nondegeneracy of the Fisher information). *Suppose Assump-*
742 *tions A4–A6 hold, and assume in addition Assumption A7. Let*

$$S^* := S(\theta^*) = \gamma^* \mu^{*\top} + U^* V^{*\top}.$$

743 *Then the Fisher information matrix*

$$I_{\Theta}(\theta^*) = \sum_{(k,i,j) \in \Omega_w} \omega_{kij} \nabla_{\theta} \eta_{kij}(\theta^*) \nabla_{\theta} \eta_{kij}(\theta^*)^{\top}$$

744 *is positive definite on the identified tangent space $T_{\theta^*}\Theta_r$; that is,*

$$\delta\theta^{\top} I_{\Theta}(\theta^*) \delta\theta > 0 \quad \text{for every } 0 \neq \delta\theta \in T_{\theta^*}\Theta_r.$$

745 *Consequently, $I_{\Theta}(\theta^*)$ is nonsingular in the local identified coordinates.*

746 *Proof of Proposition C.3.* The proof is divided into three steps.

747 **Step 1: Score-space Fisher information as a weighted Laplacian form.** Temporarily param-
 748 eterize the model by the row-centered score matrix

$$\mathcal{S} := \{S \in \mathbb{R}^{K \times N} : S1_N = 0\}.$$

749 For a fixed active triple $(k, i, j) \in \Omega_w$, recall that

$$\eta_{kij}(S) := S_{ki} - S_{kj}.$$

750 Under the binomial Bradley–Terry–Luce model, the population Fisher information in the ambient
 751 score parameterization is

$$I_S(S^*) = \sum_{(k,i,j) \in \Omega_w} \omega_{kij} \nabla_S \eta_{kij}(S^*) \nabla_S \eta_{kij}(S^*)^\top,$$

752 where $\omega_{kij} = w_{kij} p_{kij}^* (1 - p_{kij}^*) > 0$.

753 Let $H = (H_{kn}) \in \mathcal{S}$ be an arbitrary score perturbation. Then the corresponding quadratic form is

$$\langle H, I_S(S^*)H \rangle = \sum_{(k,i,j) \in \Omega_w} \omega_{kij} (H_{ki} - H_{kj})^2. \quad (\text{C.8})$$

754 Indeed, for each triple (k, i, j) , the directional derivative of $\eta_{kij}(S)$ along H is

$$D\eta_{kij}(S^*)[H] = H_{ki} - H_{kj},$$

755 which gives (C.8).

756 Now fix a judge $k \in [K]$, and write $h_k = (H_{k1}, \dots, H_{kN})^\top \in \mathbb{R}^N$. Then the k -th summand in
 757 (C.8) becomes

$$\sum_{\{i,j\} \in E_k^w} \omega_{kij} (h_{k,i} - h_{k,j})^2 = h_k^\top L_k h_k,$$

758 where L_k is the weighted graph Laplacian of the active graph $G_k^w = ([N], E_k^w)$ with edge weights
 759 ω_{kij} . Therefore

$$\langle H, I_S(S^*)H \rangle = \sum_{k=1}^K h_k^\top L_k h_k. \quad (\text{C.9})$$

760 **Step 2: Judge-wise graph connectivity implies positivity on $\mathcal{S}$.** By Assumption A5, each active
 761 graph G_k^w is connected. Because L_k is the weighted Laplacian of the connected graph G_k^w with
 762 strictly positive edge weights, for every $x \in \mathbb{R}^N$,

$$x^\top L_k x = \sum_{\{i,j\} \in E_k^w} \omega_{kij} (x_i - x_j)^2 \geq 0.$$

763 Hence $x \in \ker(L_k)$ implies $x^\top L_k x = 0$, so $x_i = x_j$ for every edge $\{i, j\} \in E_k^w$. Since G_k^w is
 764 connected, x must be constant on all vertices, that is, $x = c1_N$. Conversely, $L_k 1_N = 0$. Therefore,

$$\ker(L_k) = \text{span}\{1_N\}.$$

765 Hence, for each k ,

$$h_k^\top L_k h_k = 0 \iff h_k = c_k 1_N \text{ for some } c_k \in \mathbb{R}.$$

766 But $H \in \mathcal{S}$ means every row of H is centered, i.e.

$$H1_N = 0.$$

767 Equivalently,

$$1_N^\top h_k = 0 \quad \text{for all } k \in [K].$$

768 If $h_k = c_k 1_N$, then $0 = 1_N^\top h_k = N c_k$, so $c_k = 0$, and therefore $h_k = 0$. Since this holds for every
 769 k , we conclude that

$$\langle H, I_S(S^*)H \rangle = 0 \implies H = 0.$$

770 Thus $I_S(S^*)$ is positive definite on $\mathcal{S}$:

$$\langle H, I_S(S^*)H \rangle > 0 \quad \forall 0 \neq H \in \mathcal{S}. \quad (\text{C.10})$$

771 **Step 3: Pulling back positivity to the structured parameter space Θ_r .** Let $\delta\theta \in T_{\theta^*}\Theta_r$ be an
 772 arbitrary tangent perturbation, and define the induced score perturbation

$$H := D\Phi(\theta^*)[\delta\theta] \in \mathcal{S}.$$

773 Since

$$\Phi(\theta) = \gamma\mu^\top + UV^\top,$$

774 the first-order variation of $S(\theta)$ at θ^* is

$$H = \delta\gamma\mu^{*\top} + \gamma^*\delta\mu^\top + \delta U V^{*\top} + U^*\delta V^\top. \quad (\text{C.11})$$

775 For each active triple $(k, i, j) \in \Omega_w$, the directional derivative of the structured logit difference
 776 satisfies

$$D\eta_{kij}(\theta^*)[\delta\theta] = H_{ki} - H_{kj}.$$

777 Equivalently,

$$\nabla_{\theta}\eta_{kij}(\theta^*)^\top\delta\theta = H_{ki} - H_{kj}.$$

778 Therefore,

$$\begin{aligned} \delta\theta^\top I_{\Theta}(\theta^*)\delta\theta &= \sum_{(k,i,j) \in \Omega_w} \omega_{kij} (\nabla_{\theta}\eta_{kij}(\theta^*)^\top\delta\theta)^2 \\ &= \sum_{(k,i,j) \in \Omega_w} \omega_{kij} (H_{ki} - H_{kj})^2 \\ &= \langle H, I_{\mathcal{S}}(S^*)H \rangle. \end{aligned} \quad (\text{C.12})$$

779 Now suppose

$$\delta\theta^\top I_{\Theta}(\theta^*)\delta\theta = 0.$$

780 Then (C.12) and (C.10) imply that $H = 0$. That is,

$$D\Phi(\theta^*)[\delta\theta] = 0.$$

781 By Assumption A7, the differential $D\Phi(\theta^*)$ is injective on $T_{\theta^*}\Theta_r$, hence $\delta\theta = 0$.

782 We have thus shown that

$$\delta\theta^\top I_{\Theta}(\theta^*)\delta\theta > 0 \quad \forall 0 \neq \delta\theta \in T_{\theta^*}\Theta_r,$$

783 which proves positive definiteness of $I_{\Theta}(\theta^*)$ on the identified tangent space. Therefore $I_{\Theta}(\theta^*)$ is
 784 nonsingular in any local identified coordinate system. $\square$

785 **Remark 3** (Interpretation of the two-layer argument). The proposition separates the source of non-
 786 degeneracy into two logically distinct parts.

787 Unlike in ordinary BTL models, graph connectivity alone is not sufficient for Fisher nondegeneracy
 788 in the structured latent-factor parameterization, due to parameter-level identifiability from local reg-
 789 ularity of the structured map.. It must be combined with local injectivity of the decomposition map
 790 reflected in Assumption A7.

791 **Remark 4** (Relation to the pairwise-comparison literature). The proof above is inspired by re-
 792 cent asymptotic-inference theory for pairwise-comparison models, where the Fisher information is
 793 represented as a graph-Laplacian quadratic form and its nondegeneracy is deduced from graph con-
 794 nectivity after quotienting out the additive invariance of pairwise scores [6]. The present argument
 795 extends that idea to the heterogeneous judge-aware model by treating the structured parameter space
 796 Θ_r as a smooth submanifold of the row-centered score space $\mathcal{S}$, and then pulling back the ambient
 797 Fisher information through the differential of the map $\Phi(\theta) = \gamma\mu^\top + UV^\top$.

798 In other words, the graph structure controls nondegeneracy in the ambient score geometry, whereas
 799 the constraint geometry of Θ_r controls whether such nondegeneracy is retained after passing to the
 800 latent parameters.

C.5 Partial sampling scheme

Let $\mathcal{G} := ([N], \mathcal{E})$ denote the pooled graph where $\mathcal{E} := \{\{i, j\} : (k, i, j) \in \Omega_w \text{ for some } k\}$ denotes the edge set. If the graph belonging to a judge, say, $\mathcal{G}_k$ is not connected, then the sampling is considered partial. For example, a judge may compare Item 1, 2, 3 and 4, 5, 6, but the graphs of 1–2–3 and 4–5–6 are separated. Under such a scenario, identification becomes the major obstacle to statistical analysis. The judge may assign higher scores to all 1, 2, and 3, and lower scores to all 4, 5, and 6 without breaking Equation (2) or altering the likelihood. Therefore, assumptions should be made to rule out such ambiguity: the true parameter shall lie in a region where such perturbation must make it impossible for S to still admit the structured representation (3) with unchanged r .

We define the structured score manifold

$$\mathcal{M}_r := \{S = \gamma\mu^\top + UV^\top : (\gamma, \mu, U, V) \in \Theta_r\} \cap \mathcal{S},$$

and let $\mathcal{T}^* := T_{S^*}\mathcal{M}_r$ be its tangent space at S^* to characterize the potential local perturbations on S^* .

Assumption A8 (Identification under partial overlap). Every nonzero local perturbation of the structured score matrix changes at least one observed pairwise log-odds difference. Equivalently, if $H \in \mathcal{T}^*$ satisfies

$$H_{ki} - H_{kj} = 0 \quad \text{for all } (k, i, j) \in \Omega_w,$$

then we must have $H = 0$.

On the other hand, we assume connectivity on the pooled graph to ensure that there are shared nodes (items) that connect the disconnected parts of each judge, so information is expected to be aligned with the help of these connecting nodes.

Assumption A9 (Connectivity of the pooled graph). The pooled graph $\mathcal{G} = ([N], \mathcal{E})$ with edge set $\mathcal{E} := \{\{i, j\} : (k, i, j) \in \Omega_w \text{ for some } k\}$ is connected.

Remark 5 (How overlap helps). The point of Assumptions A8 and A9 is that overlap lets different judges' local comparisons be aligned through the shared low-rank structure. For example, suppose judge 1 compares items (1, 2) and judge 2 compares items (2, 3), but neither judge compares all three pairs. Item 2 then serves as an overlap point linking the two local pieces of information. Assumption A8 dictates that, after imposing the structured model, such overlaps are rich enough that one cannot change the latent score matrix in a nontrivial way without altering at least one observed comparison.

Assumption A10 (Local compact parameter chart). There exists a compact set $\Theta_{r,0} \subset \Theta_r$ such that θ^* lies in its relative interior after fixing the local anchor signs of the columns of U and V . And the population structured criterion

$$m(\theta) := \sum_{(k,i,j) \in \Omega_w} w_{kij} \left[-p_{kij}^* \eta_{kij}(\theta) + \log(1 + \exp\{\eta_{kij}(\theta)\}) \right],$$

has θ^* as its unique minimizer over $\Theta_{r,0}$.

This assumption replaces the coercivity established in Lemma C.2 under Assumption A5 (full sampling setting). As the identification constraint becomes obscured for the entire space $\mathcal{S}$, this assumption, together with Assumption A10, dictates that even though the likelihood may be flat along component-wise shifts of an unrestricted row, those shifts should either be infeasible within the fixed-rank structured model or break the $\Theta_{r,0}$ boundary that we set.

By the standard asymptotic theory of MLE, the asymptotic properties of $\hat{\theta}$ of Theorem 3 still hold under Assumptions A4, A6 and A8–A9.

In practice, however, we need to avoid this setting due to the difficulty in verifying Assumptions A8 and A10, unless the sample size is really limited.

D Rank selection

D.1 Rank selection criteria

We offer three ways to select the rank r of our model following Section 5.4.

Cross Validation. In experiments with dense comparison graphs, we can mask a validation subset of the pairwise comparisons $\Omega_{\text{val}} \subset \Omega$. For each candidate rank r , we estimate the parameters $\hat{\theta}_r$ on the training set and select the r that minimizes the negative log-likelihood on Ω_{val} . This guards against overfitting by measuring generalizability and is hence suitable for empirical practices.

Spectral Residual Analysis. According to (5), estimation of the preference residuals $UV^\top$ as a whole does not depend on r . Therefore, as a straightforward diagnostic, r can be selected by analyzing the spectral decay of $UV^\top$, which can be estimated by fitting the consensus-only model ($S_0 = \gamma\mu^\top$) and compute $S - S_0$. A scree plot of SVD of this residual matrix can help locate the optimal r which typically corresponds to the elbow of this plot, reflecting the dominant r disagreement aspects. This approach is more statistically heuristic and offers interpretation to our model.

Information-theoretic approach: BIC. We also propose BIC method for hyperparameter selection, which is statistically guaranteed to consistently estimate the true rank r^* of the underlying judge preference matrix $(UV^\top)^*$. Note that the BIC for a candidate rank r is defined as

$$\text{BIC}(r) = 2\mathcal{L}_n(\hat{\theta}_r) + d_r \log n, \quad (\text{D.1})$$

where $\mathcal{L}_n(\hat{\theta}_r)$ is the minimized negative log-likelihood under rank r , $d_r := r(K + N - r - 3)$ is the effective degrees of freedom (having accounted for the identifiability constraints).

D.2 Proof of Theorem 4

We provide the formal proof for Theorem 4. The goal is to show that minimizing the Bayesian Information Criterion (BIC) consistently selects the true heterogeneity rank r^* as the total number of comparisons $n \rightarrow \infty$ while the comparison graph Ω remains fixed. Note that the sample log-likelihood is $-\mathcal{L}_n(\theta)$ throughout the paper.

Proof of Theorem 4. Let $\mathbb{P}^*$ denote the true data-generating distribution parameterized by $\theta^* \in \Theta_{r^*}$, where Θ_r is the constrained parameter space corresponding to rank r (adhering to the identifiability constraints in Proposition 1). To prove $\lim_{n \rightarrow \infty} \mathbb{P}(r^* = \arg \min_r \text{BIC}(r)) = 1$, we must show that for any $r \neq r^*$, $\mathbb{P}(\text{BIC}(r) - \text{BIC}(r^*) > 0) \rightarrow 1$. We partition the analysis into two cases: the under-parameterized regime ($r < r^*$) and the over-parameterized regime ($r > r^*$).

Case 1: Under-parameterized regime ($r < r^*$). When $r < r^*$, the candidate model space Θ_r does not contain the true parameter θ^* . Because the BTL model is a strictly concave Generalized Linear Model (GLM) mapping bounded logits to probabilities, and our identifiability constraints make the parameterization injective, θ^* cannot be perfectly approximated by any $\theta \in \Theta_r$.

Following the argument of Section C.2 (especially (C.4)), by the properties of the KL divergence, the maximum expected log-likelihood under the restricted space Θ_r is strictly less than under the true parameter:

$$\delta_r := M(\theta^*) - \sup_{\theta \in \Theta_r} M(\theta) > 0$$

By the Uniform Law of Large Numbers (since Θ is compact and the BTL log-likelihood is continuous and bounded), we have $\frac{1}{n}\mathcal{L}_n(\hat{\theta}_r) \xrightarrow{\mathbb{P}} \sup_{\theta \in \Theta_r} M(\theta)$ and $\frac{1}{n}\mathcal{L}_n(\hat{\theta}_{r^*}) \xrightarrow{\mathbb{P}} M(\theta^*)$.

The difference in BIC is

$$\begin{aligned} \text{BIC}(r) - \text{BIC}(r^*) &= -2 \left(\mathcal{L}_n(\hat{\theta}_{r^*}) - \mathcal{L}_n(\hat{\theta}_r) \right) + (d_r - d_{r^*}) \log(n) \\ &= -2n \left(\frac{1}{n} \mathcal{L}_n(\hat{\theta}_{r^*}) - \frac{1}{n} \mathcal{L}_n(\hat{\theta}_r) \right) + (d_r - d_{r^*}) \log(n). \end{aligned} \quad (\text{D.2})$$

880 Since the term inside the first parenthesis converges in probability to $\delta_r > 0$, the likelihood differ-
 881 ence scales as $\mathcal{O}_p(n)$. But the penalty difference $(d_r - d_{r^*}) \log(n)$ only scales as $\mathcal{O}(\log n)$, making
 882 the linear growth of the likelihood deficit dominate. Thus, for any $r < r^*$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(\text{BIC}(r) - \text{BIC}(r^*) > 0) = 1.$$

883 **Case 2: Over-parameterized regime ($r > r^*$)** When $r > r^*$, the candidate model space Θ_r
 884 strictly nests the true model space Θ_{r^*} in terms of closure (i.e., $\overline{\Theta_{r^*}} \subset \overline{\Theta_r}$). Therefore, $\theta^* \in \overline{\Theta_r}$,
 885 and the model can be viewed as correctly specified but over-parameterized.

886 Because the true model is nested within the larger model, we invoke theories on Likelihood Ratio
 887 (LR) test. Under the identifiability constraints (Section 4.3) and assumptions required by asymptotic
 888 normality, the Fisher Information Matrix evaluated at θ^* is non-singular. Note that under the null
 889 hypothesis that the true parameter lies in the restricted space Θ_{r^*} , the log-likelihood ratio statistic
 890 may not converge in law to a chi-squared distribution, as the true parameter θ^* essentially lies on the
 891 boundary of Θ_r . However, $-2 \left(\mathcal{L}_n(\hat{\theta}_r) - \mathcal{L}_n(\hat{\theta}_{r^*}) \right)$ is still $\mathcal{O}_p(1)$ by theories on the boundary be-
 892 haviour of LR [7, Theorem 3] and tests on reduced-rank regression [8, Theorem 3.2]. Therefore, the
 893 maximum improvement in log-likelihood gained by fitting $r - r^*$ spurious components is bounded in
 894 probability. But $r > r^*$ leads to $d_r > d_{r^*}$ and consequently the penalty term $(d_r - d_{r^*}) \log(n) \rightarrow \infty$
 895 as $n \rightarrow \infty$. Thus, for any $r > r^*$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\text{BIC}(r) - \text{BIC}(r^*) > 0) = 1.$$

896 Combining Case 1 and Case 2, the true rank r^* strictly minimizes the BIC with probability tending
 897 to 1 as the sample size $n \rightarrow \infty$, i.e., $\lim_{n \rightarrow \infty} \mathbb{P}(\arg \min_r \text{BIC}(r) = r^*) = 1$. $\square$

898 E Detailed Experimental Instructions

899 This appendix records the intended local execution protocol so the empirical section can be repro-
900 duced and extended without changing the main-paper narrative.

901 E.1 Repository structure

902 All empirical runs should use the ranking repository created for this project. The relevant files are:

- 903 • `src/ranking_judge/`: runner, providers, cache, and prompt parsing code;
- 904 • `configs/providers/`: free-tier and paid provider presets;
- 905 • `configs/experiments/`: reusable experiment definitions;
- 906 • `data/examples/`: toy and near-tie JSONL examples;
- 907 • `runs/`: per-experiment outputs.

908 E.2 Environment setup

- 909 1. Create the local conda environment from `environment.yml`.
- 910 2. Install the package in editable mode with `python -m pip install -e ..`
- 911 3. Set exactly one provider key for the intended run, for example `OPENROUTER_API_KEY` or
912 `GEMINI_API_KEY`.
- 913 4. Confirm that the target provider config points to the intended model and base URL before run-
914 ning large batches.

915 E.3 Data preparation

916 **Synthetic block.** Sample a consensus vector μ , nonnegative judge reliabilities γ , and low-rank
917 factors U and V . Form $S = \gamma\mu^\top + UV^\top$, then generate repeated pairwise outcomes from Equa-
918 tion (4). The simulation should vary both the dispersion of γ and the magnitude of $UV^\top$ so that
919 reliability and structured disagreement can be disentangled.

920 **Prompt-sensitivity block.** Use MT-Bench-style pairwise rows with fields `example_id`,
921 `question`, `answer_a`, and `answer_b`. Keep candidate answers fixed while varying only the judging
922 prompt and answer order.

923 **Near-tie block.** Construct a curated near-tie dataset by editing strong answers for length, for-
924 matting, or mild phrasing changes while preserving semantic content. Every example should be
925 manually checked to ensure that multiple verdicts remain plausible.

926 E.4 Prompt protocol

- 927 1. Use the MT-Bench pairwise prompt as the baseline condition.
- 928 2. Add a tie-aware judge-aware prompt that explicitly instructs the judge not to reward verbosity
929 and to return `[[C]]` when the answers are substantively indistinguishable.
- 930 3. Run each pair in both original and swapped answer order.
- 931 4. For reasoning-heavy subsets, keep a reference-based prompt variant available even if the main
932 benchmark does not assume ground truth.

933 E.5 Judge-panel protocol

- 934 1. Start with one free-tier judge configuration for low-cost iteration.
- 935 2. Add at least one additional judge model when panel evaluation begins.
- 936 3. Keep prompts, candidate answers, and default temperature fixed across judges.
- 937 4. Estimate both pooled rankings and judge-specific disagreement patterns from the same compar-
938 ison set.

939 **E.6 Caching, retries, and parse safety**

940 Every request must be hashed from the exact provider name, model, prompt template, messages, and
941 sampling parameters. Identical requests should hit the SQLite cache instead of the API. Transient
942 failures should be retried with bounded exponential backoff, while configuration failures such as
943 missing API keys should fail immediately and be written as explicit error records. Parse failures are
944 not to be dropped; the raw response text must be retained for later analysis.

945 **E.7 Expected run artifacts**

946 Each experiment should emit:

- 947 • `records.jsonl`, one row per request with parsed verdict and raw response;
- 948 • `summary.json`, machine-readable counts and metadata;
- 949 • `summary.md`, a short human-readable summary table.

950 These artifacts are the minimal record needed for reproducibility and later paper tables.

951 **E.8 Example command sequence**

```
952 python -m ranking_judge run \  
953     --provider configs/providers/openrouter_free.json \  
954     --experiment configs/experiments/mtbench_pairwise.json  
955  
956 python -m ranking_judge run \  
957     --provider configs/providers/gemini_free.json \  
958     --experiment configs/experiments/prompt_sensitivity.json  
959  
960 python -m ranking_judge run \  
961     --provider configs/providers/openrouter_free.json \  
962     --provider configs/providers/openai_compatible_paid.example.json \  
963     --experiment configs/experiments/panel_judging.json
```

964 For a paper-ready empirical version, the final appendix should also record the exact model identi-
965 fiers, prompt text revisions, date ranges of API usage, and any manually curated exclusions from the
966 near-tie benchmark.

References

- [1] Hamed Karimi, Julie Nutini, and Mark Schmidt. Linear convergence of gradient and proximal-gradient methods under the polyak-łojasiewicz condition. In *Joint European conference on machine learning and knowledge discovery in databases*, pages 795–811. Springer, 2016.
- [2] Boris Teodorovich Polyak. Gradient methods for minimizing functionals. *Zhurnal vychislitel'noi matematiki i matematicheskoi fiziki*, 3(4):643–653, 1963.
- [3] Hedy Attouch, Jérôme Bolte, Patrick Redont, and Antoine Soubeyran. Proximal alternating minimization and projection methods for nonconvex problems: An approach based on the kurdyka-łojasiewicz inequality. *Mathematics of operations research*, 35(2):438–457, 2010.
- [4] Angelika Bunse-Gerstner, Ralph Byers, Volker Mehrmann, and Nancy K Nichols. Numerical computation of an analytic singular value decomposition of a matrix valued function. *Numerische Mathematik*, 60(1):1–39, 1991.
- [5] Emanuel Malveti, Gunther Dirr, Frederik vom Ende, and Thomas Schulte-Herbrüggen. Analytic, differentiable and measurable diagonalizations in symmetric lie algebras. *arXiv preprint arXiv:2212.00713*, 2022.
- [6] Ruijian Han, Wenlu Tang, and Yiming Xu. Statistical inference for pairwise comparison models. *arXiv preprint arXiv:2401.08463v3*, 2025.
- [7] Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987.
- [8] Jean-Marc Robin and Richard J. Smith. Tests of rank. *Econometric Theory*, 16(2):151–175, 2000.